Density Estimation

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Density estimation

- Estimating a probability distribution from data
- True distribution of data: P_{data}
- Estimated distribution: $P_{estimate}$

Density estimation

Parametric
(Essentially
curve fitting)

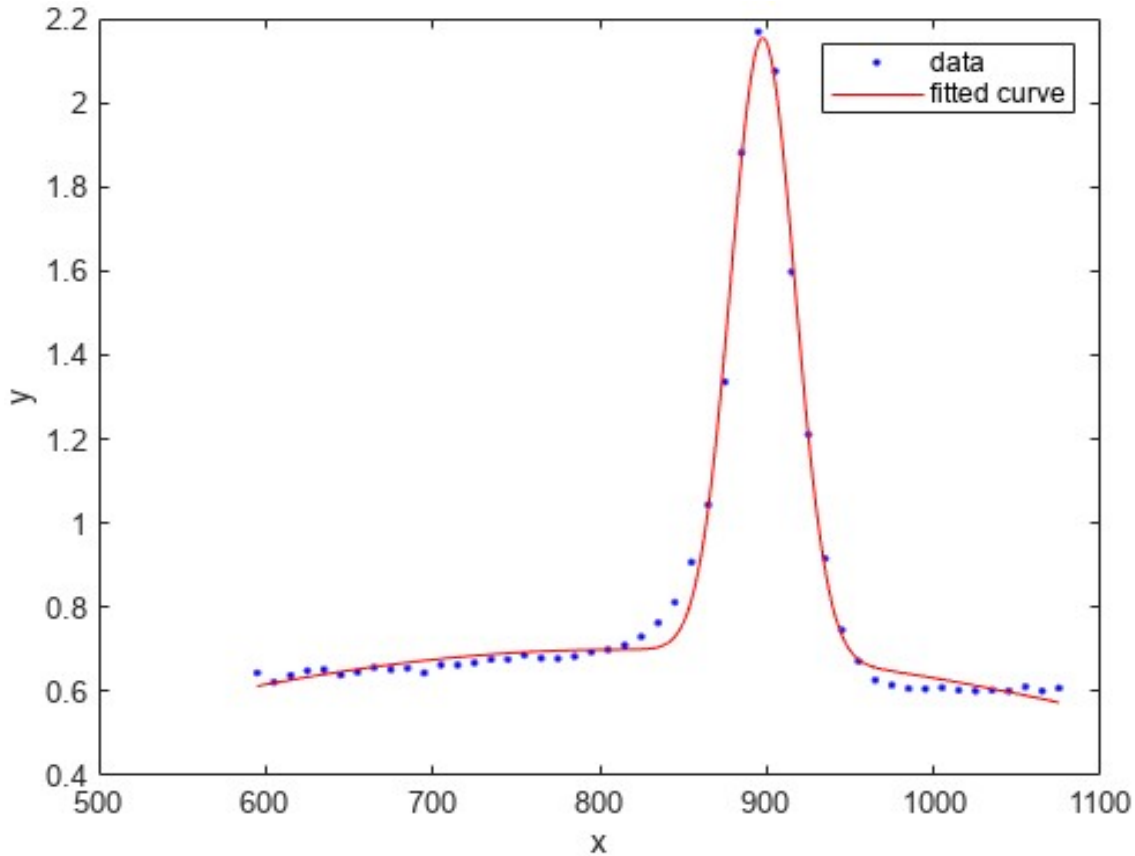
Non-parametric

- Histogram Estimator
- Kernel Density Estimator
- Naive Estimator
- KNN estimator

(Nearest Neighbour Estimator)

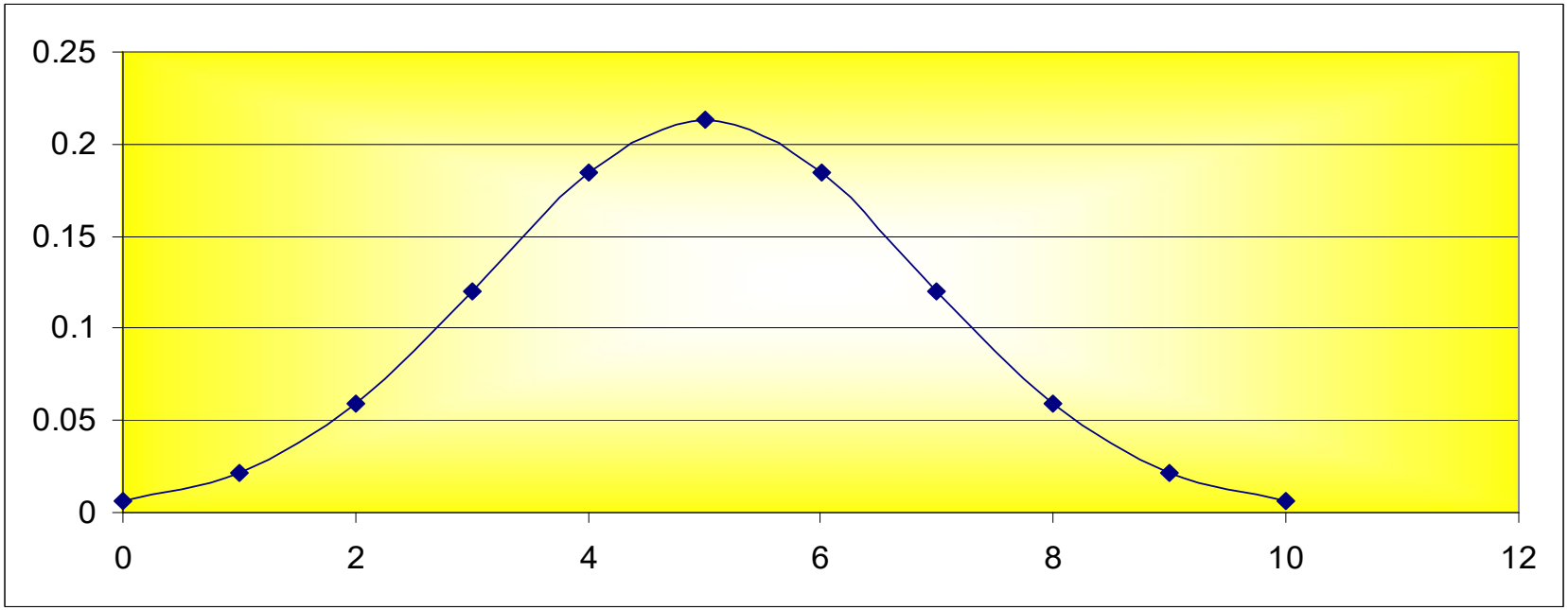
Parametric estimation

- Assume the population i.e. data has distributed in certain fashion
- Some body assures the distribution



Nothing but curve fitting

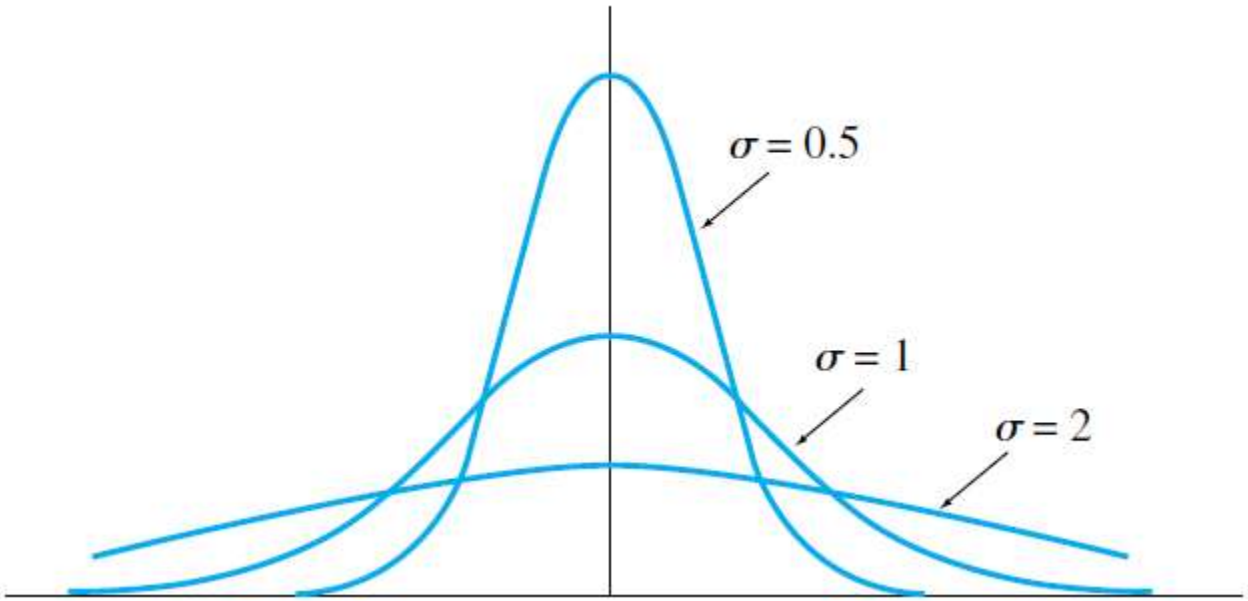
Gaussian distribution – mean and standard deviation



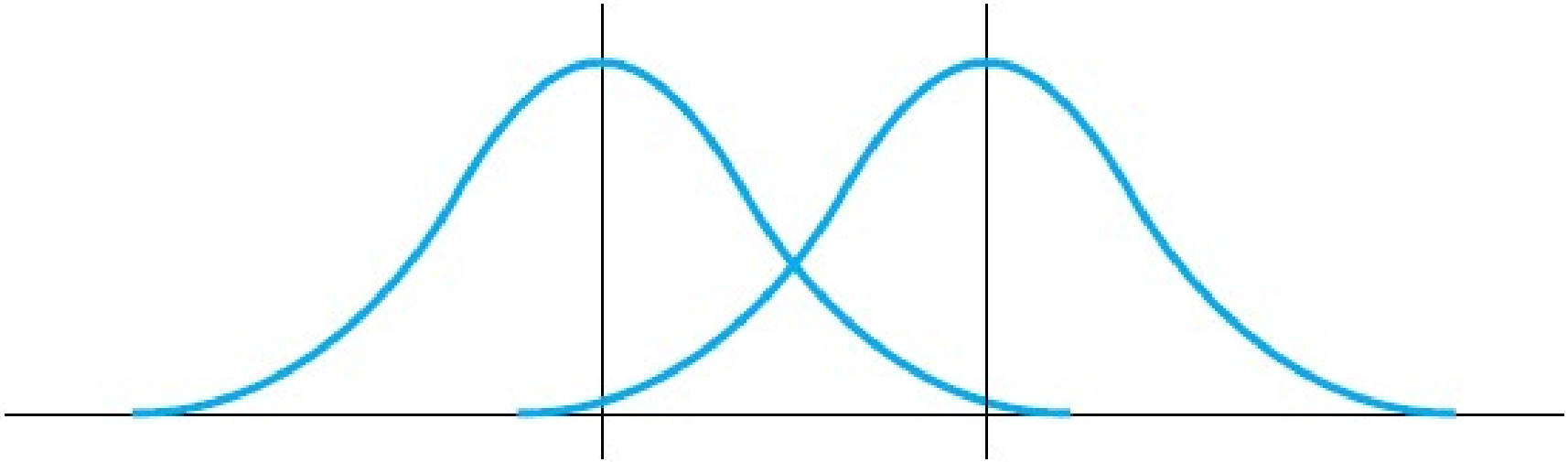
$$\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Defined from $-\infty$ to $+\infty$

Same mean – different standard deviations



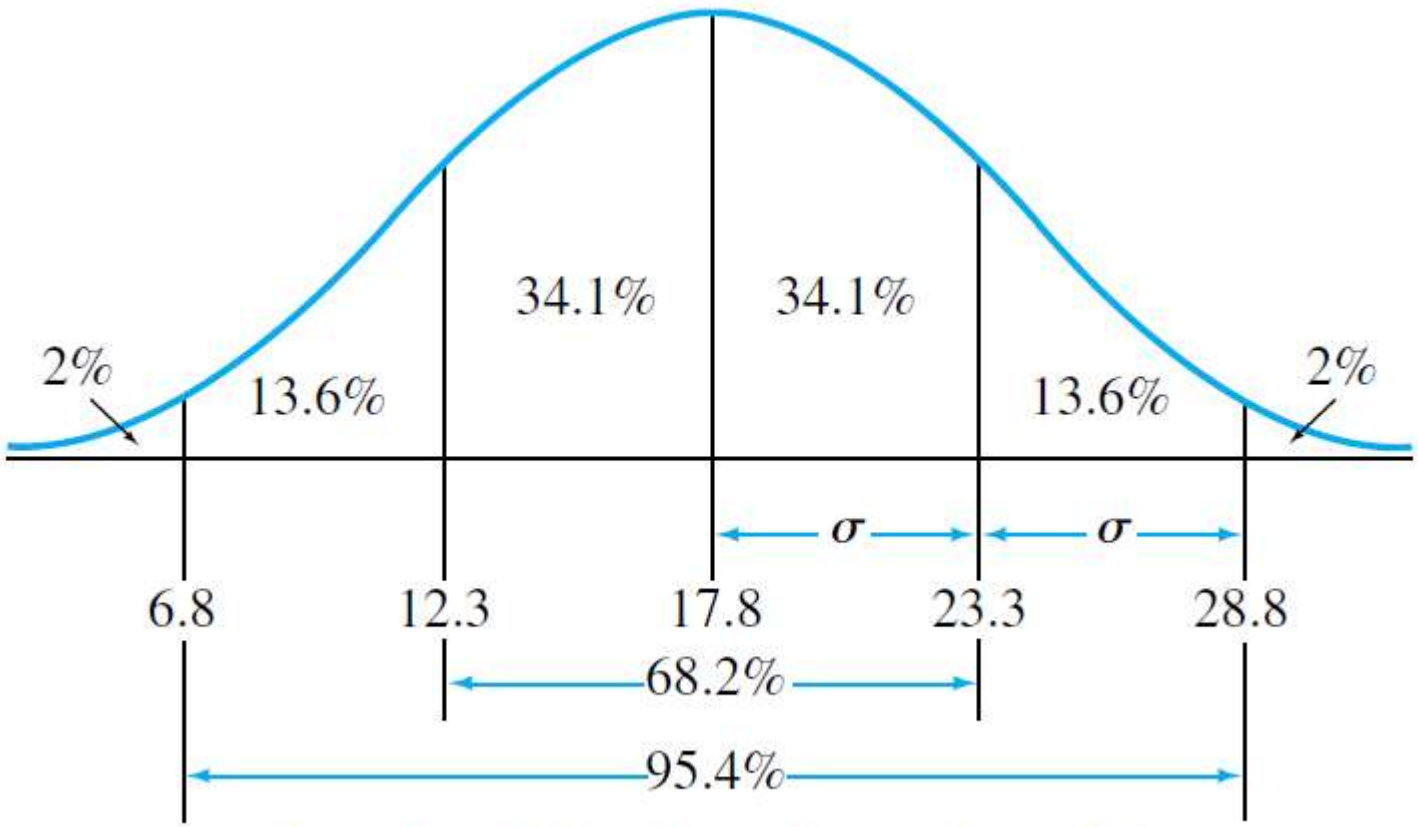
Same standard deviation – different means



Standard normal deviation

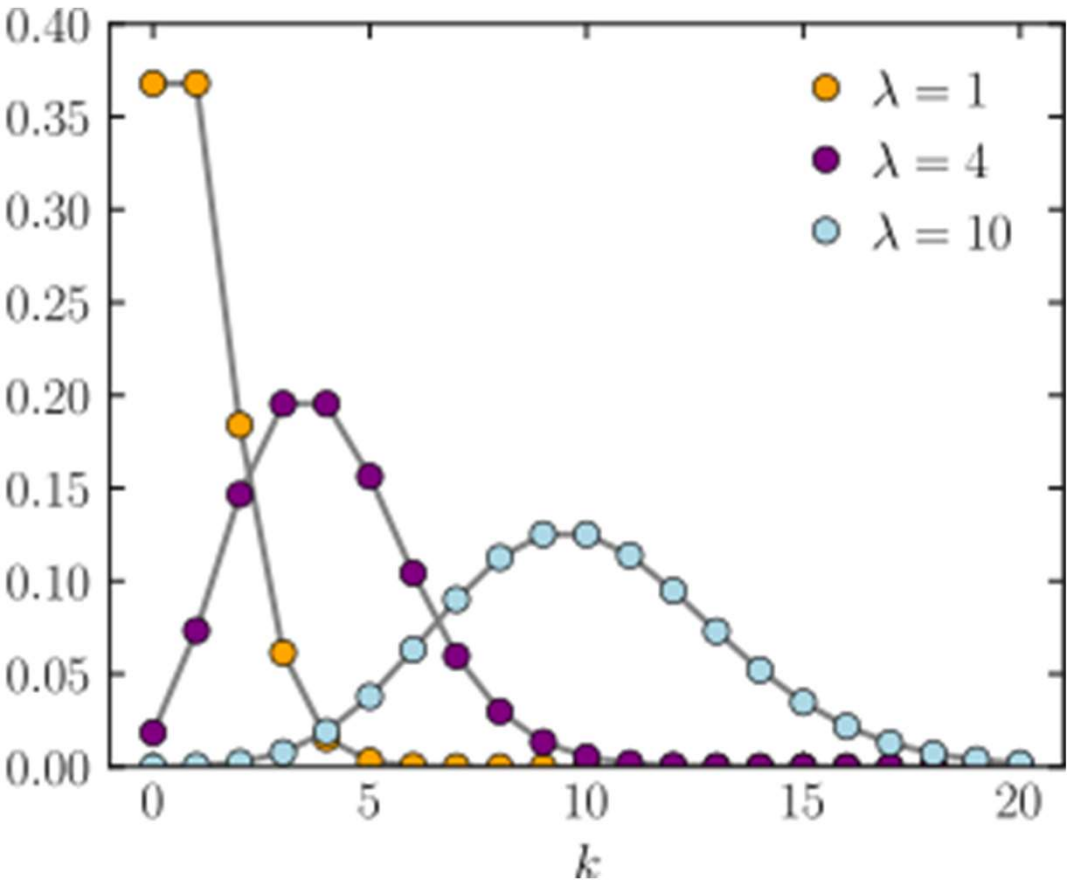
- Make, $\sigma = 1$
- Area under the curve becomes 1
- can be used as probability measure

Area under the normal curve



Normal curve with $\mu = 17.8$ and $\sigma = 5.5$

Poisson's distribution



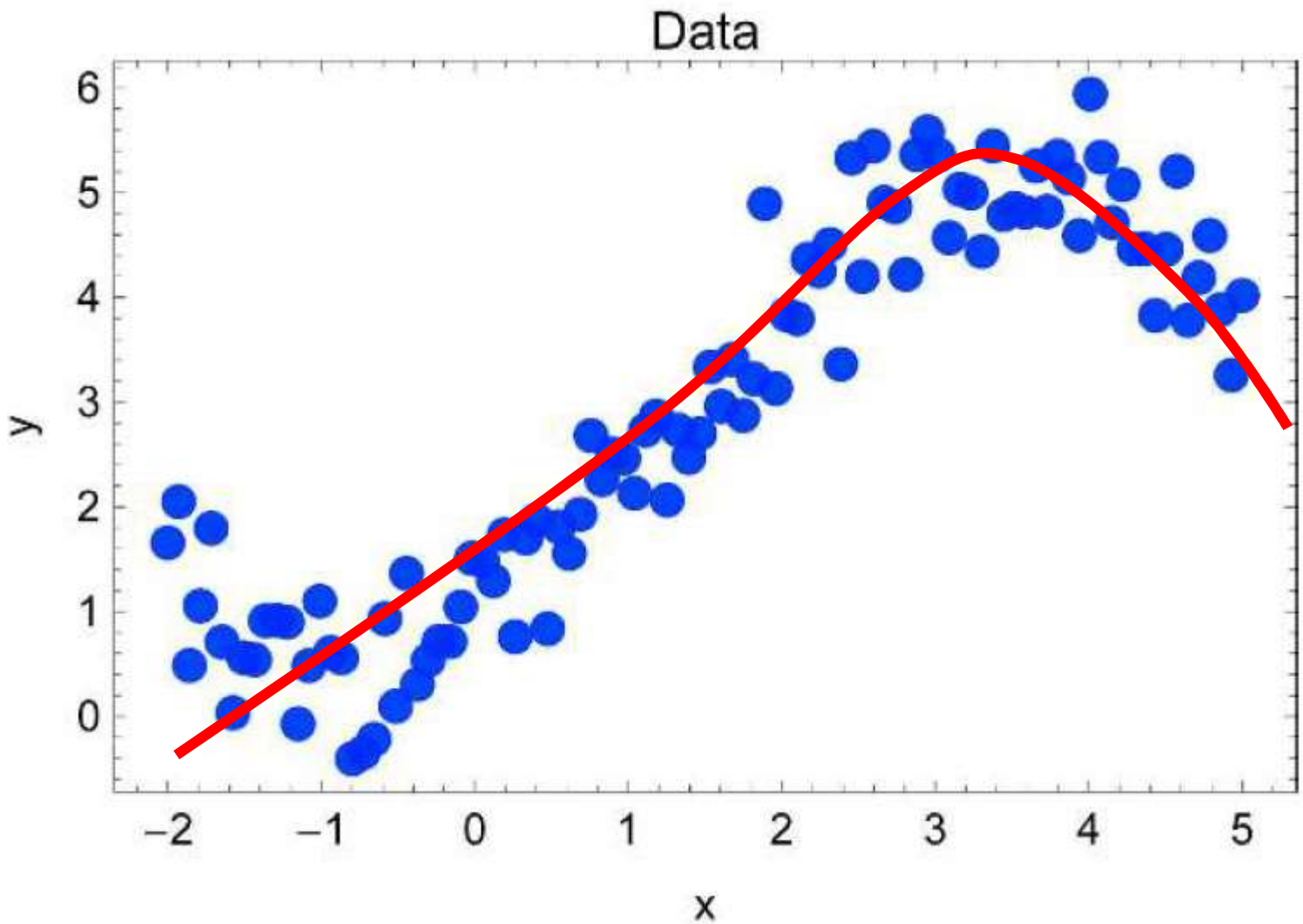
$$f(x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

Estimate the distribution parameters

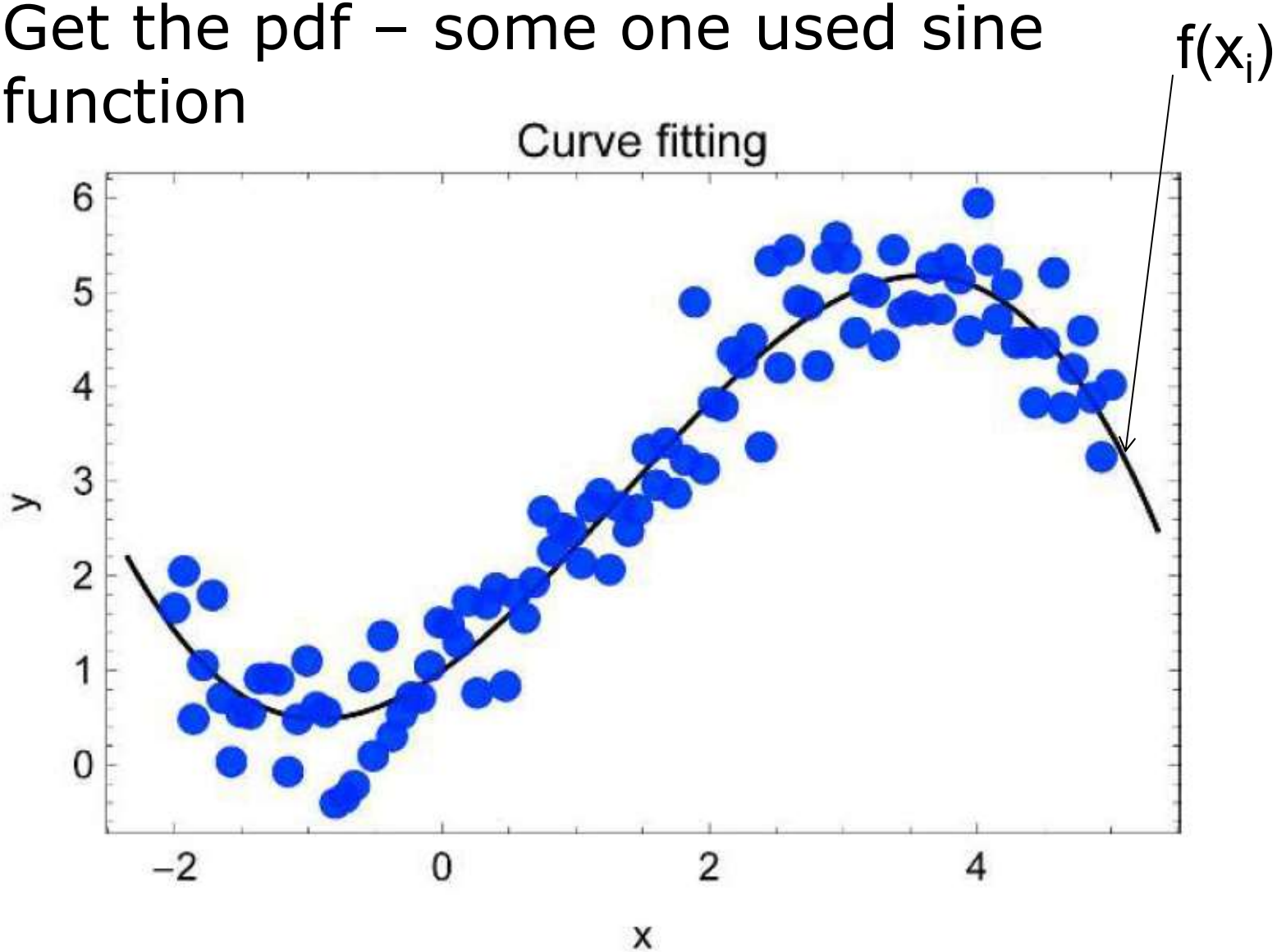
- After guessing distribution, estimate the distribution parameters

- Gaussian
  - Estimate μ and σ
- Poisson
  - Estimate λ

- Get the pdf – someone used Gaussian



- Get the pdf – some one used sine function

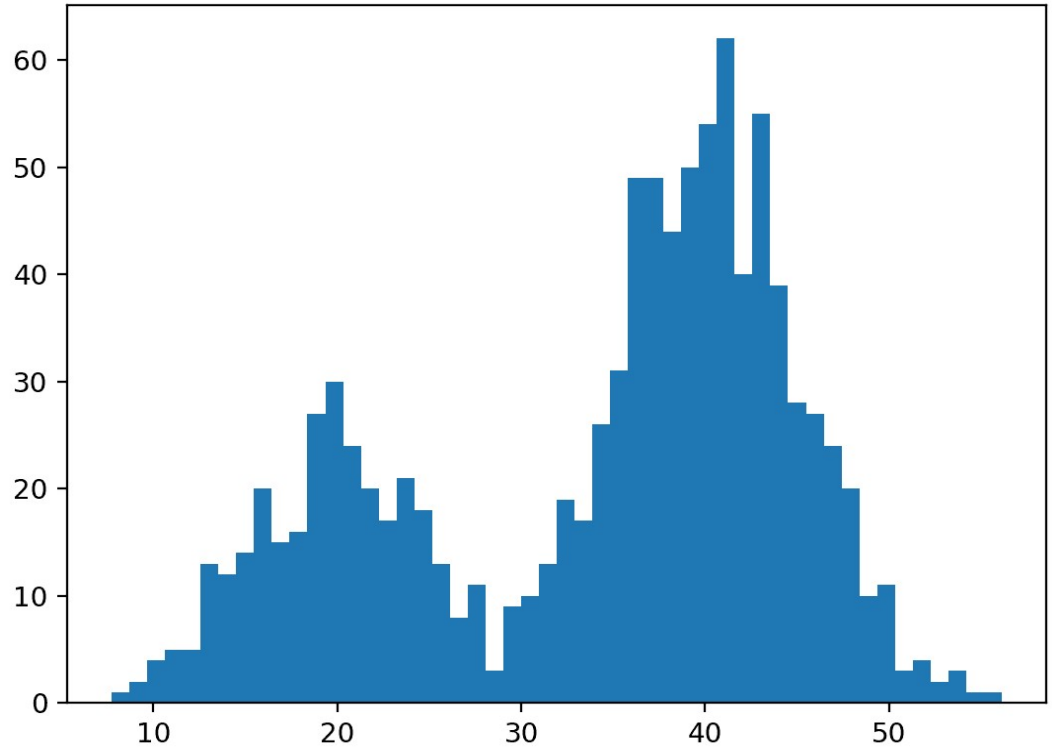


Nonparametric probability density estimation

- To fit a model to the arbitrary distribution of the data
  - Histogram density estimation
  - Kernel density estimation

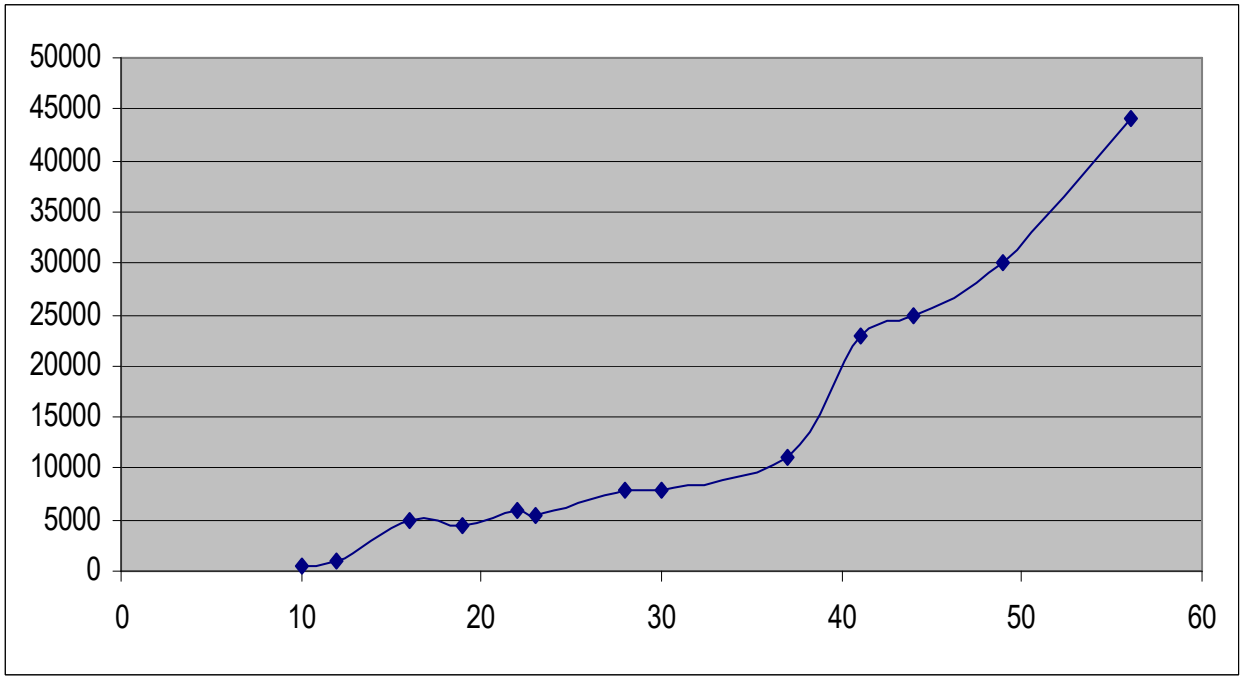
Non-parametric estimation

- If we are not able to guess the distribution
  - May be we have two peaks or more than two peaks
- Kernel density estimation



Data visualization

AGE	Ca - **deficiency**
9	500
12	1000
16	5000
19	4500
22	6000
23	5500
28	8000
30	7800
37	11000
41	23000
44	25000
49	30000
56	44000



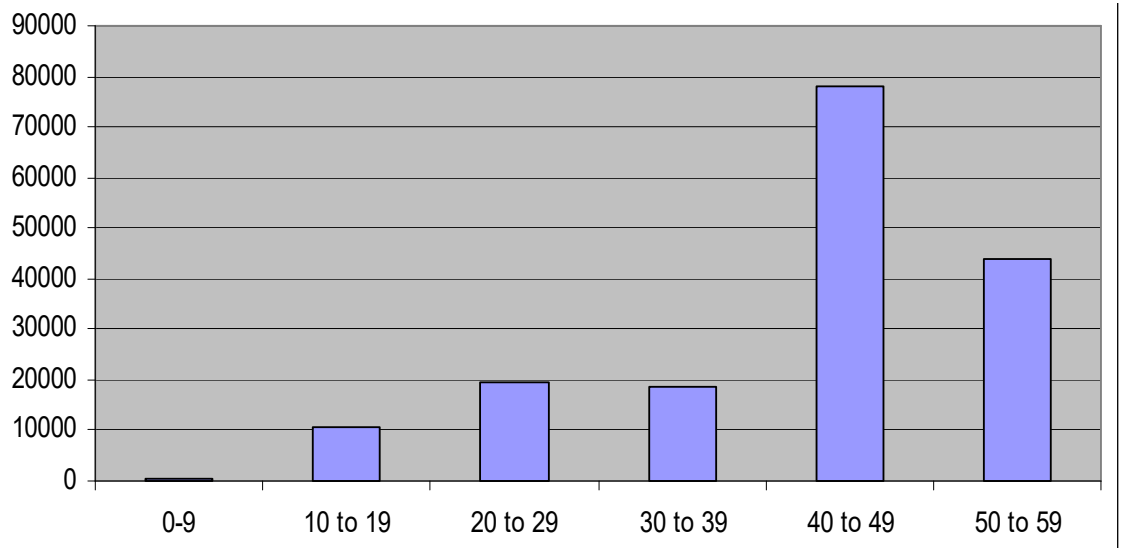
Conclusion from the previous slide

- As age increases people more prone for Calcium deficiency

If you are > 50 years

"Drink Bournvita"

9	500
12	1000
16	5000
19	4500
22	6000
23	5500
28	8000
30	7800
37	11000
41	23000
44	25000
49	30000
56	44000

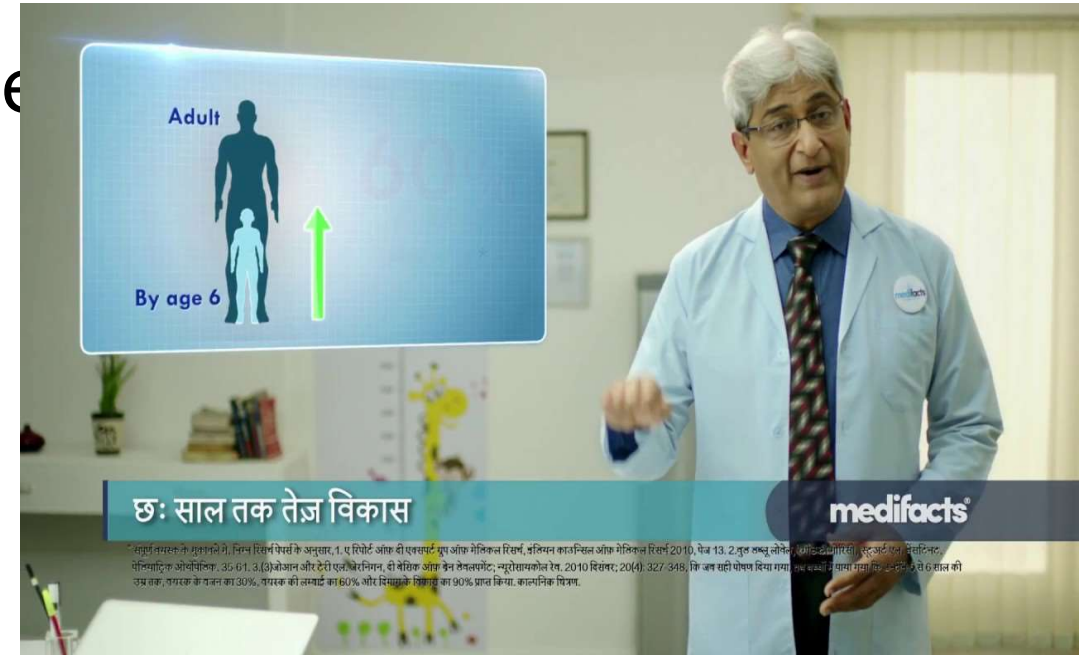


0-9	500
10 to 19	10500
20 to 29	19500
30 to 39	18800
40 to 49	78000
50 to 59	44000

Histogram

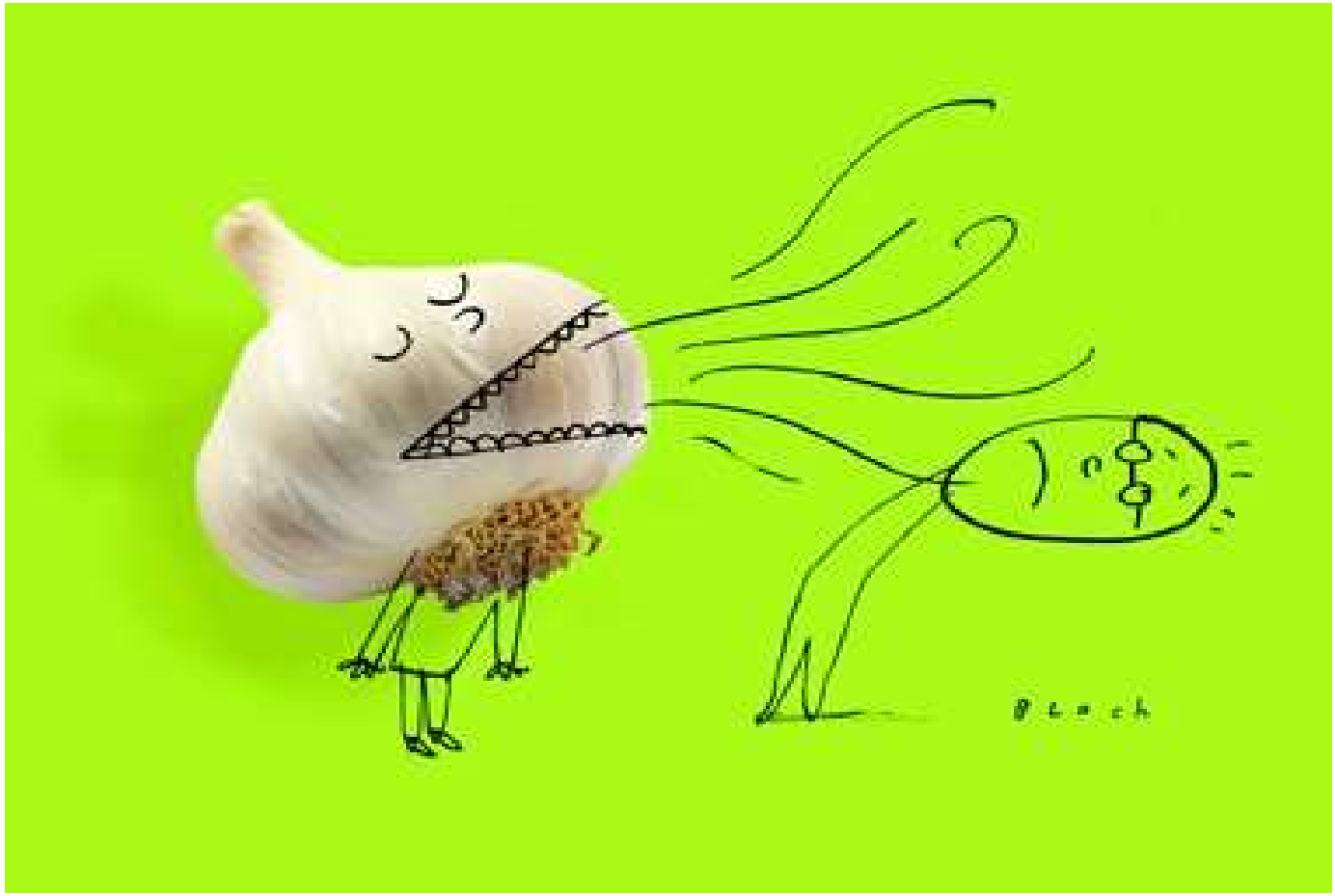
Horlicks Advertisement: from the previous slide

•People in 40-49 age group more prone for Calcium deficiency

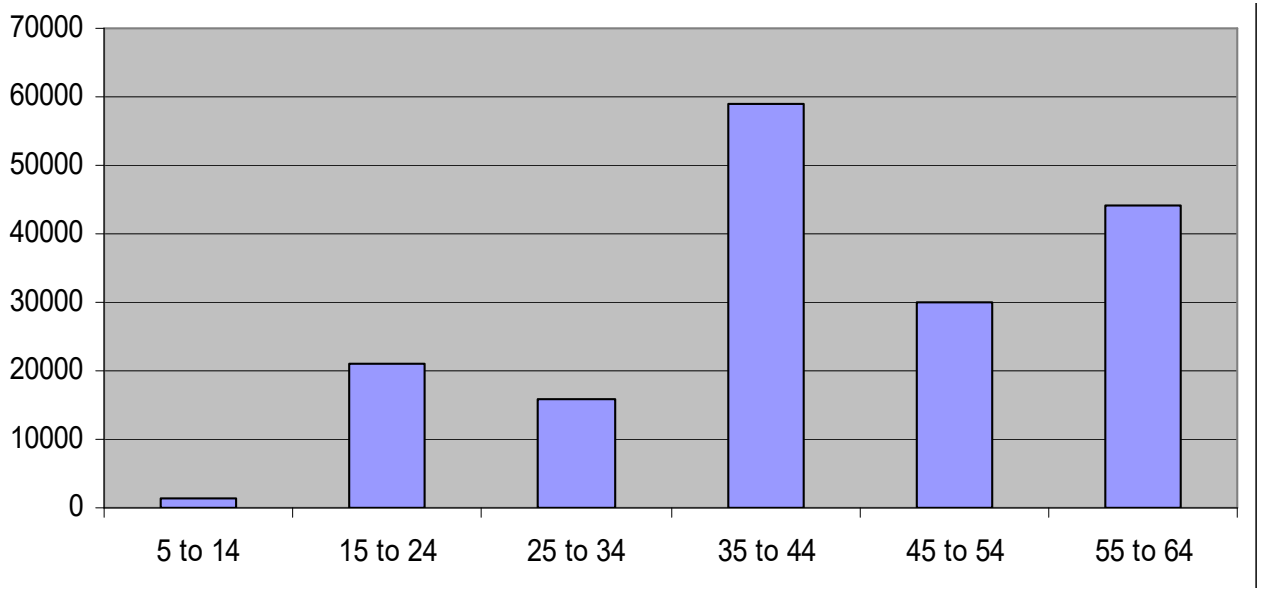


If you are 40-49
“Drink Horlicks”

Let us start a company “Garlicks”



9	500
12	1000
16	5000
19	4500
22	6000
23	5500
28	8000
30	7800
37	11000
41	23000
44	25000
49	30000
56	44000

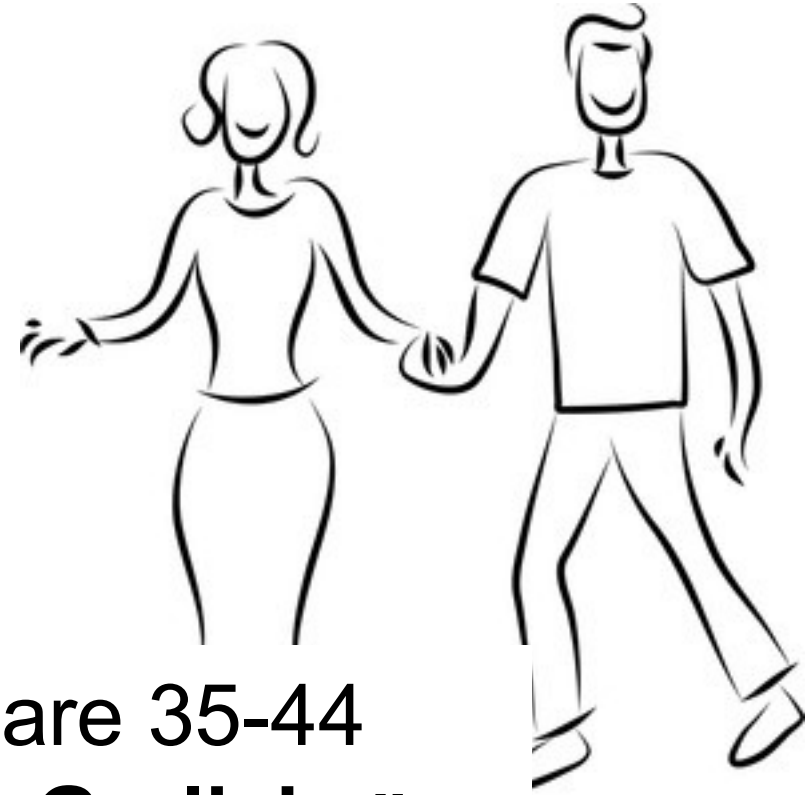


5 to 14	1500
15 to 24	21000
25 to 34	15800
35 to 44	59000
45 to 54	30000
55 to 64	44000

Histogram

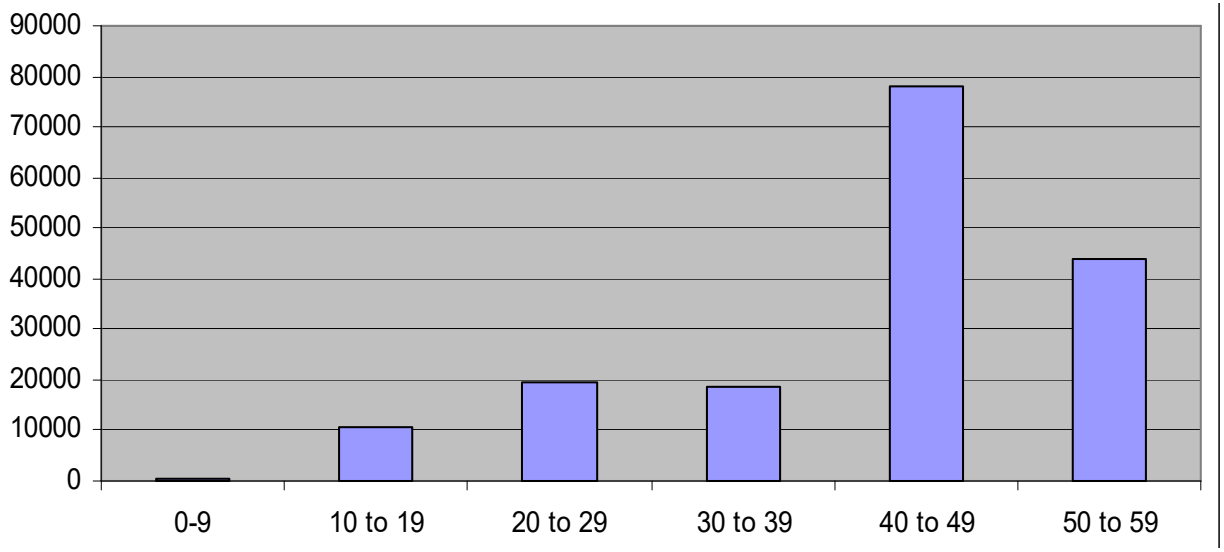
Garlicks advertisement: from the previous slide

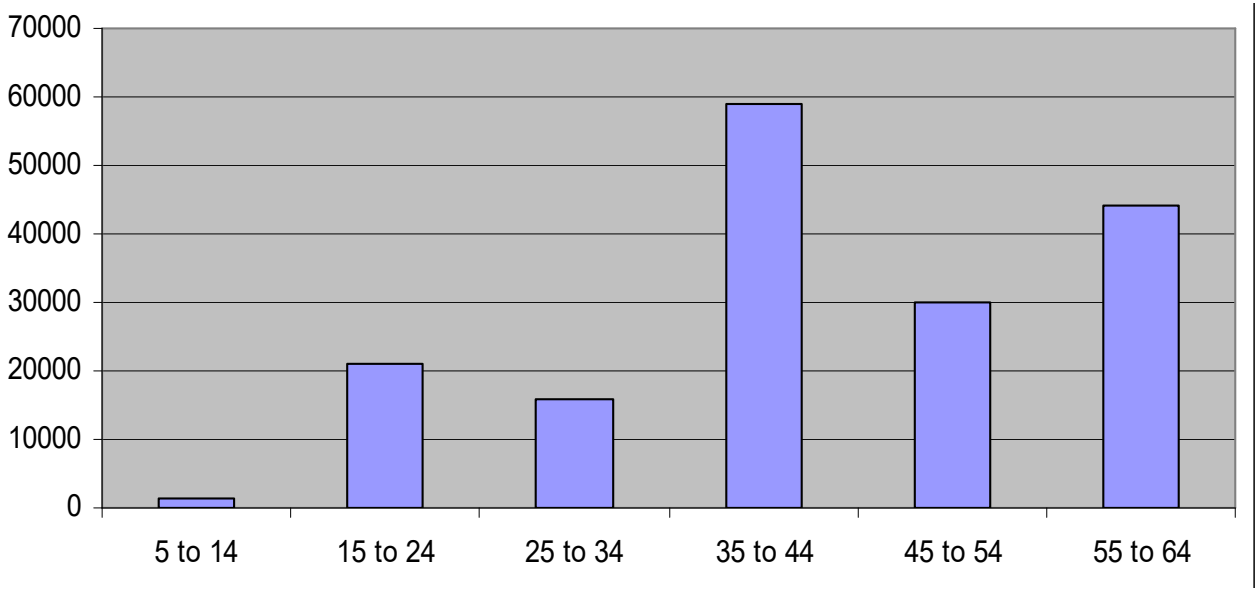
- A recent study shows that people in 35-44 age group also more prone for Calcium deficiency



If you are 35-44
"Drink Garlicks"

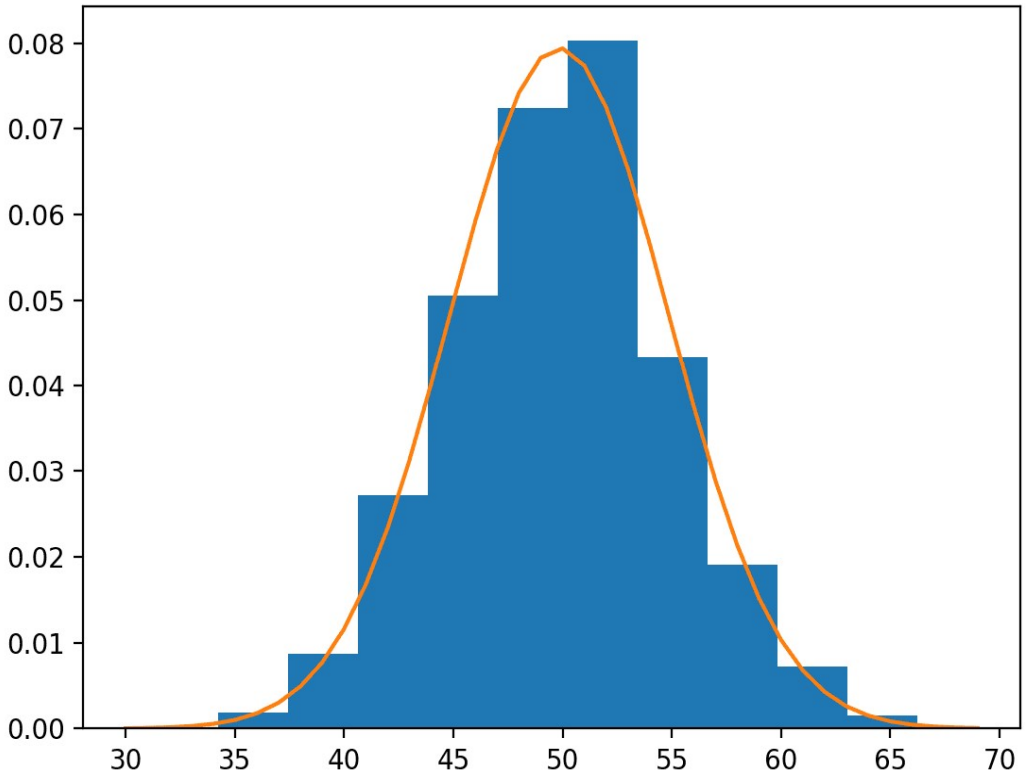
9	500
12	1000
16	5000
19	4500
22	6000
23	5500
28	8000
30	7800
37	11000
41	23000
44	25000
49	30000
56	44000





Which is correct?

Histogram density estimation



- Generate histogram from the given data
- Look @ the shape
- Try to guess the distributions
- Popular distributions
  - Gaussian
  - Poisson
  - Uniform

Histogram - problems

- Histogram shape depends on the bin width
- Change the bin width – shape also changes
- Suppose bin width is constant. Is histogram unique? – NO
  - 0-2; 2-4; 4-6; … are bins & bin width = 2
  - -1-+1; 1-3; 3-5; … are bins & bin width=2
- Bin origin is another problem

Bin width

- Smoothing parameter
  - A smaller binwidth leads to a relatively jagged histogram
  - A larger binwidth results in a smoother looking histogram

- The bin width acts as a "smoothing parameter."
  - Smaller width = less smoothing (more detail, more noise)
  - Larger width = more smoothing (less detail, less noise)

Summary Table

Bin Width	Shape of Histogram	Pro/Con
Small (narrow)	Jagged, more detail/noise	Shows fine details, but may exaggerate noise
Large (wide)	Smoother, less detail	Hides noise, but may oversimplify real structure

In Practice

- Choosing bin width is a balance: too small, and you see only noise; too large, and you lose meaningful shape.
- There's no single "correct" bin width for all data—often, it's helpful to try several settings to understand what's real.

What is Bin Width?
Bin width is the size of each interval or "bin" on the x-axis into which data points are grouped in a histogram.
The choice of bin width directly controls how much data is "summed" in each bar.

What is a Bin Edge?
In a histogram, bin edges are the boundaries that separate the intervals (bins) into which your data points are grouped.
For example, if your bins are 0–2, 2–4, 4–6, then 0, 2, 4, 6 are the edges.

Bin edge

- Sensitivity of the histogram to the placement of the bin edge
- Average shifted histogram:
  - Averages several histograms based on shifts of the bin edges

Average Shifted Histogram (ASH) – The Solution
How Does ASH Work?
ASH is a technique designed to reduce the dependence of the histogram on bin edge placement.
Instead of making just one histogram:
Construct several histograms, each with the same bin width but starting the first bin at slightly different positions (shifts).
For example: Start bins at 0, then at 0.2, then at 0.4, etc.
Average the counts (heights) at each x value across all these shifted histograms.
What Does This Achieve?
By averaging, ASH produces a smoother, less sensitive estimate of the data's density.
It lessens the impact of any one arbitrary edge position, making the histogram a more reliable reflection of the data's true shape.

Summary Table

Concept	Explanation
Bin edge sensitivity	The shape/appearance of a histogram changes if you shift where bins start
Problem	Critical features can be hidden/exaggerated just due to arbitrary bin edge placement
Average shifted histogram	Create several histograms with shifted bins, then average them to reduce edge sensitivity

In short:
The placement of histogram bin edges can strongly affect the appearance and interpretation of your data. The *average shifted histogram* solves this by averaging over multiple shifted versions, making your histogram more stable and meaningful.

Sample No.	Value
1	-2.1
2	-1.3
3	-0.4
4	1.9
5	5.1
6	6.2

If the histogram is pdf then

- $f(x) = \text{limit } (h \to 0)\ 1/(2h).\ P(x-h < X < x+h)$
- Area under the bins should add up to 1
- Window size
  - More the window width more number of points will fall
- Weight for one point in the window
  - Inversely proportional to the number of points (n)
  - Also inversely proportional to window size
    - Bigger the window size many points will fall within window i.e. lesser weight for one point
    - Smaller the window size few points will fall within window i.e. lesser weight for one point

Weight per Point:
Inversely proportional to points per window (n), and also to window width (h).
Intuition: If many points in a bin, each “matters less” to the total.

- **PDF via Histogram:**
  - The probability density function at x, **f(x)**, is defined (roughly) by counting how many points fall in a tiny window ("2h" wide) around x, dividing by window width, and normalizing as h→0.
  - **Formula:**
    $f(x) = \lim_{h\to 0} \frac{1}{2h} P(x-h < X < x+h)$
    - This formalizes "density at x."
- **Area Under Histogram = 1:**
  - The pdf (when estimated via histogram) must have total area 1, just like any probability distribution.
- **Window Size (h):**
  - **Bigger window (bin):** More points fall into each bin.
  - **Smaller window:** Fewer points per bin, more "jagged."
- **Weight per Point:**
  - **Inversely proportional to points per window** (n), and also to window width (h).

- Histogram ≡ stacking boxes
  - One box width is 2h
  - One box height 1/(2hn)
- Total box width = 2h
- Total box height = N*[1/(2hn)] where N is the number of points within the window
- If there are M boxes then
  - Area of all the M boxes = 1

- Box height is the density estimate

- **Histogram as PDF Estimate:**
  - When you normalize the histogram (so that area = 1), you essentially have a crude estimate of the data's pdf.
- **Bin (Window) Size Tradeoff:**
  - **Wider bins:** Smoother, less noise, may miss detail.
  - **Narrower bins:** Reveals more detail but can be noisy/jagged.
- **Calculation of Density:**
  - For a given bin (centered at some x):

$$\text{Density} = \frac{\text{number of points in bin}}{\text{total points} \times \text{bin width}}$$

  - This ensures the area of all histogram bars sums to 1.
- **Wider windows → more points per bin but lower height; narrower windows → fewer points but more variable height.**
- **This is a "non-parametric" density estimate** u don't assume a specific distribution shape, you let the data dictate the histogram shape.

- Histogram bins can be imagined as boxes:
  - **Width = 2h:** The window spans x–h to x+h.
  - **Height = 1/(2hn):** For a bin with a single point, the "density" is 1 over (bin width × sample size).
- **If N points fall in a bin (window):**
  - The total height is N × [1/(2hn)] = N/(2hn).
- **Area of Each Bin:**
  - **width × height = 2h × (N/(2hn)) = N/n:**
    - So the fraction of area in a bin equals the fraction of data points it contains.
- **Total Area = 1:**
  - If there are M bins covering all the data, the total area sums to 1 (since the sum of fractions $N_x/n + N_y/n$... = 1).

Box Height = Density Estimate

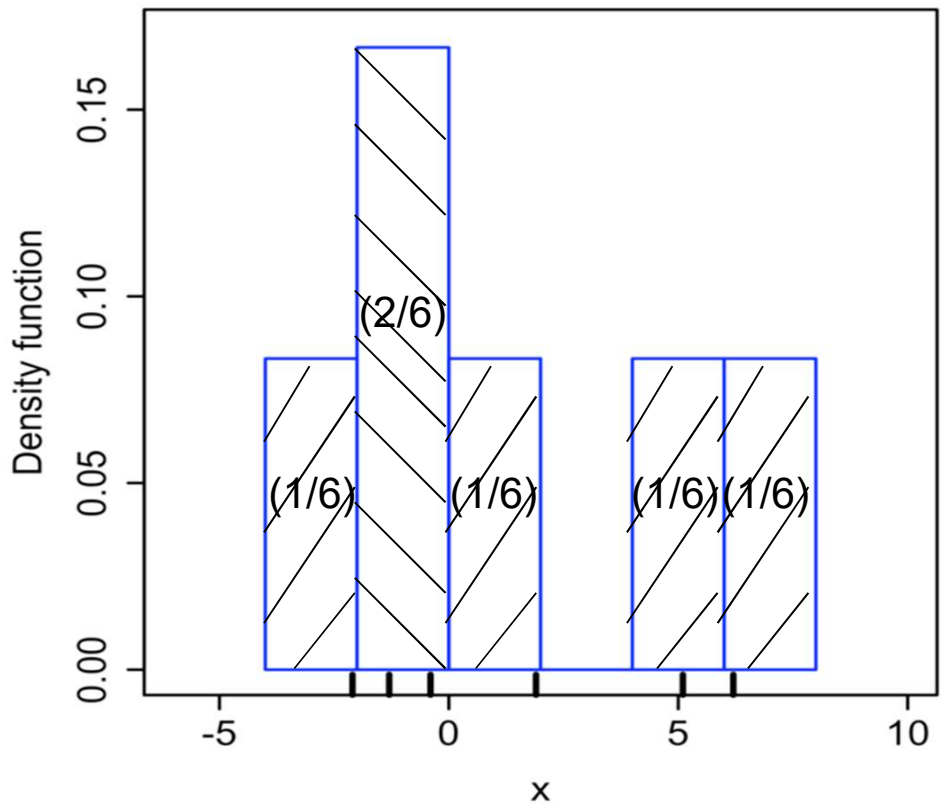
- The **height of each bar** gives you the estima... probability density for that bin.

Area of one bar=bin width×height

Histogram

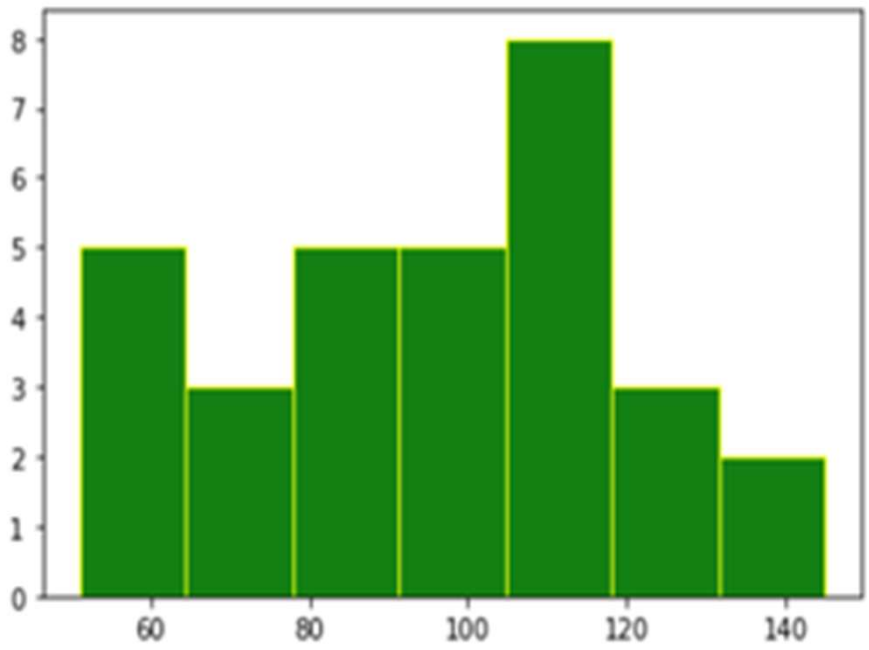
-4 to -2; -2 to 0; 0 to 2; 2 to 4; & 4 to 6

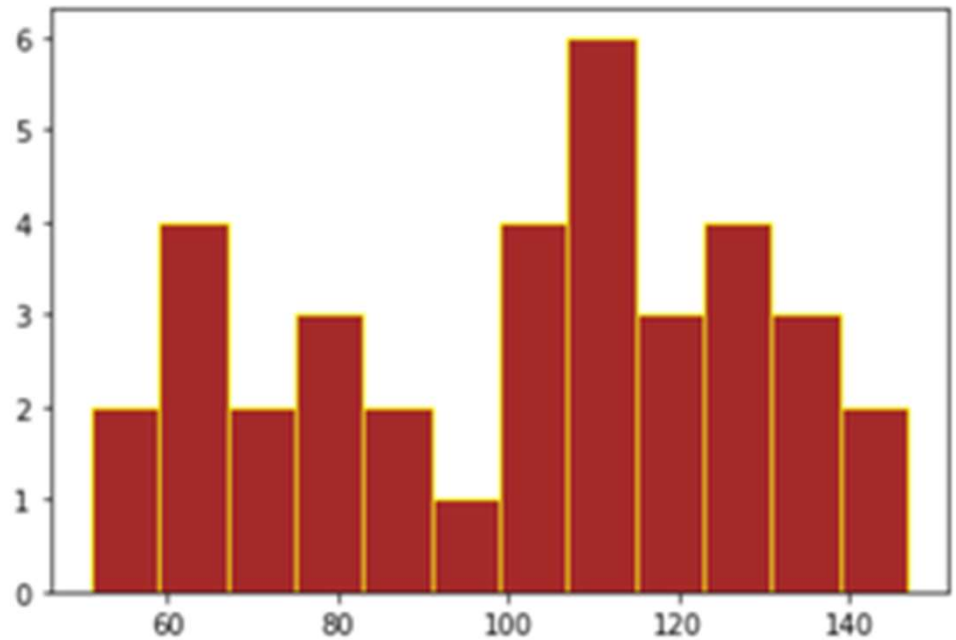
- Horizontal axis is divided into sub-intervals or bins which cover the range of the data
- E.g. 6 bins each of width 2
  - One data point falls inside this interval
  - n = 6 and 2h = 2 i.e. h = 1
  - Height = 1/(2nh) = 1/12
  - 1 point in a bin → Place 1 box of height 1/12
  - 2 points in a bin → Place two boxes (2 x 1/12)
  - And so on…
  - Area of one box = 2 x 1/12 = 1/6
  - Area of six boxes = 6x1/6 = 1



Ambiguity in histogram

- Bin width





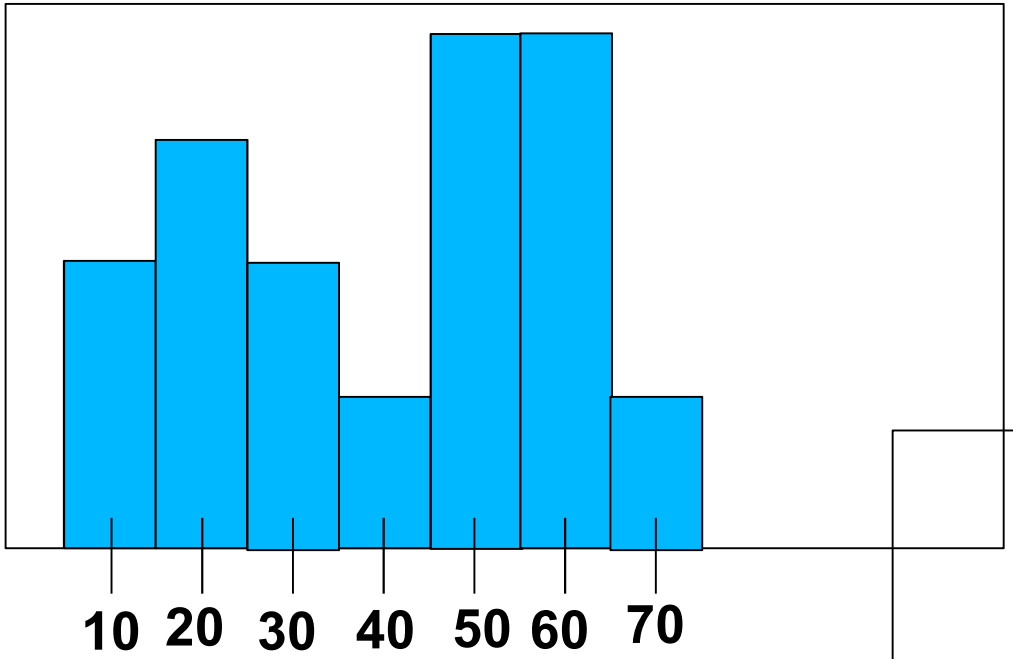
- The left histogram (green): wide bins, fewer bars, higher bars.
- The right histogram (red): narrow bins, more bars, but each is lower.

Explanation:

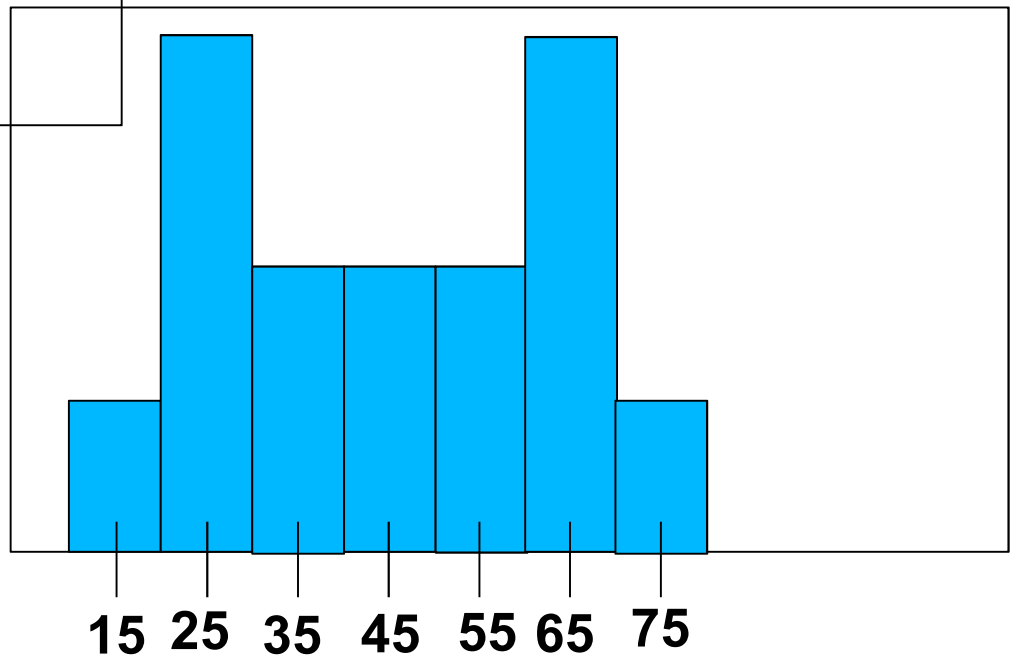
- **Bin Width = Smoothing Parameter:**
  - **Wide bins (left):** Smoother, but less detail. Small variations in data are "smoothed over."
  - **Narrow bins (right):** Jagged, shows more details and fluctuations, but can also reveal noise that isn't meaningful.
- **Ambiguity:**
  The general *visual impression* of the data distribution is different — you may conclude the underlying "shape" or modality of your data is different based just on bin width choice.

Key Point:
Histograms are not unique! Their appearance—and even your interpretation—depends heavily on the bin width you pick. There's a trade-off between detail and noise.

Ambiguity in histogram



- Bin edge



- The only thing changed: **where the first bin edge is placed** (the "bin origin").
  - Top-left starts at 10, goes 10-30-50...
  - Bottom-right starts at 15, goes 15-35-55...

Explanation:

- **Bin Edge Sensitivity:**
  Where you "start" your bins (where the first bin begins) radically shifts which data points fall into which bins. This can **hide or split clusters**, create or destroy apparent peaks, or change the visual symmetry.
- **What Is an Average Shifted Histogram?**
  To reduce this sensitivity, you can **create several histograms** with the same bin width but different starting points; then average their heights. This gives a smoother, more stable estimate.

Key Point:
Not only does the **width** of the bins matter, but so does **where they start** — another source of histogram ambiguity.

1d histogram

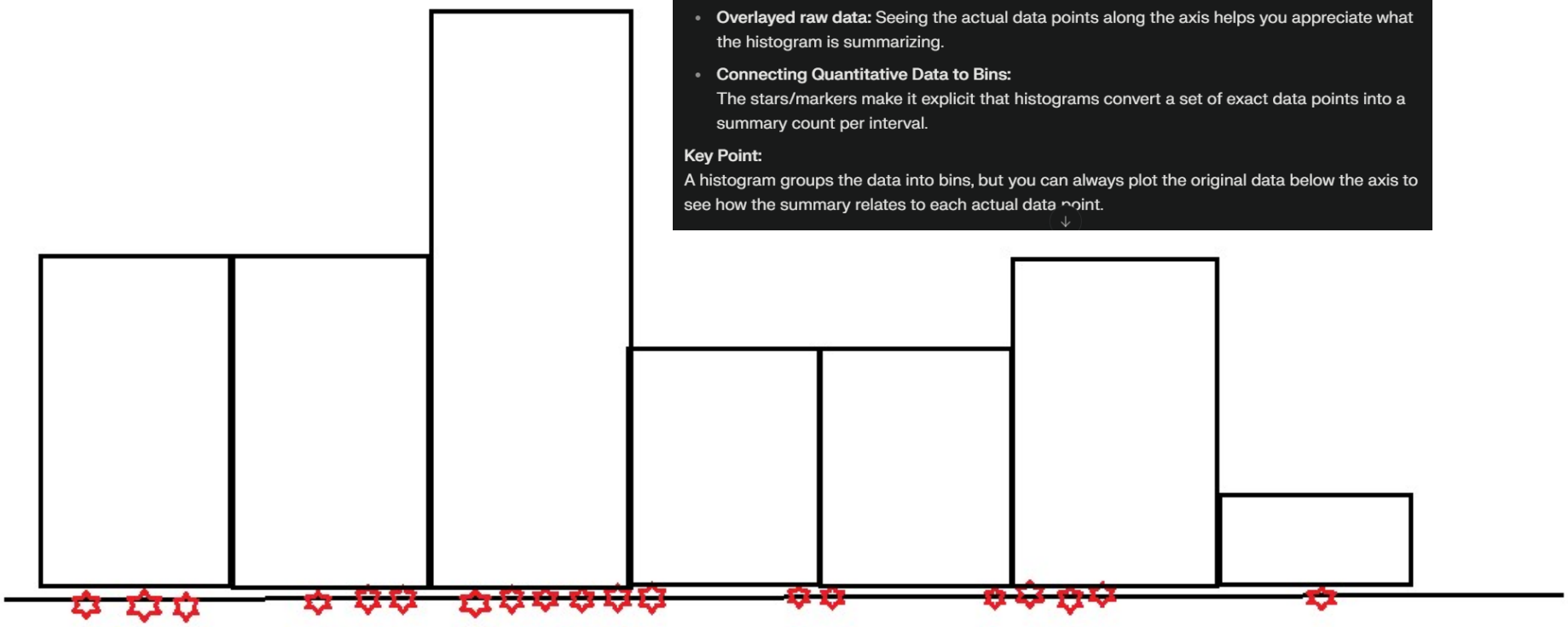
What's Shown:

- A bar-only histogram with no fill, showing intervals (bins) on the x-axis.
- Red stars/o's align along the number line at the bottom — these are the individual data points (the "raw" one-dimensional samples).

Explanation:

- **Bars:** Show how many data points fall inside each bin ("count" per interval).
- **Overlayed raw data:** Seeing the actual data points along the axis helps you appreciate what the histogram is summarizing.
- **Connecting Quantitative Data to Bins:**
  The stars/markers make it explicit that histograms convert a set of exact data points into a summary count per interval.

Key Point:
A histogram groups the data into bins, but you can always plot the original data below the axis to see how the summary relates to each actual data point.



What about 2d histogram?

- 1d - data
- Assume entire data range is divided into 5 bins

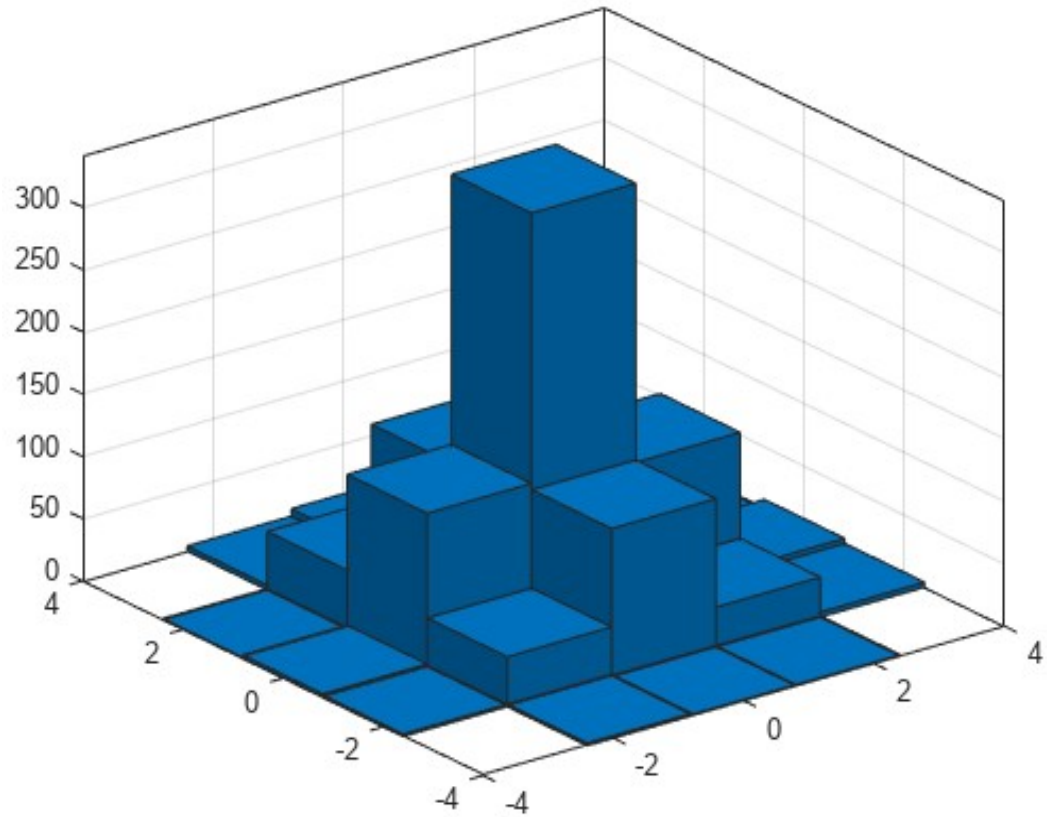
- 2d - data
- We need 25 bins

- Comparison between a **1D histogram** (bars, left) and a **2D histogram** (3D plot of "heights" in a grid, right).
- Text explains: dividing a 1D range into 5 bins means 5 bars; dividing both X and Y axes into 5 bins each makes a grid of **25** "bins" (5 × 5).

Explanation:

- **2D Histogram:**
  The idea generalizes: if you have two variables and bin *each* axis, your "bins" are now 2D rectangles (cells in a grid). The "height" above each cell is the **count** of data points falling in that rectangle.
- **Bin Explosion:**
  Binning in more dimensions means the total number of bins grows *very* fast (exponentially): 5 in 1D, 25 in 2D, 125 in 3D, and so on.
- **Challenges in High Dimensions:**
  Too many bins can mean most bins are empty unless you have lots of data ("curse of dimensionality").

Key Point:
Histograms can work in higher dimensions, but bin choices matter even more, and data becomes sparse in many bins quickly.



Limitations of histogram

- Not smooth
- Sudden changes in magnitude
- Shape of the histogram depends on the bin positions
- No. of grid cells needed to partition the input space of x
- Increases exponentially with data dimension
  - E.g. 1d-data: N^1
  - 2d-data: N^2
  - 3d-data: N^3
  - Nd-data: N^n

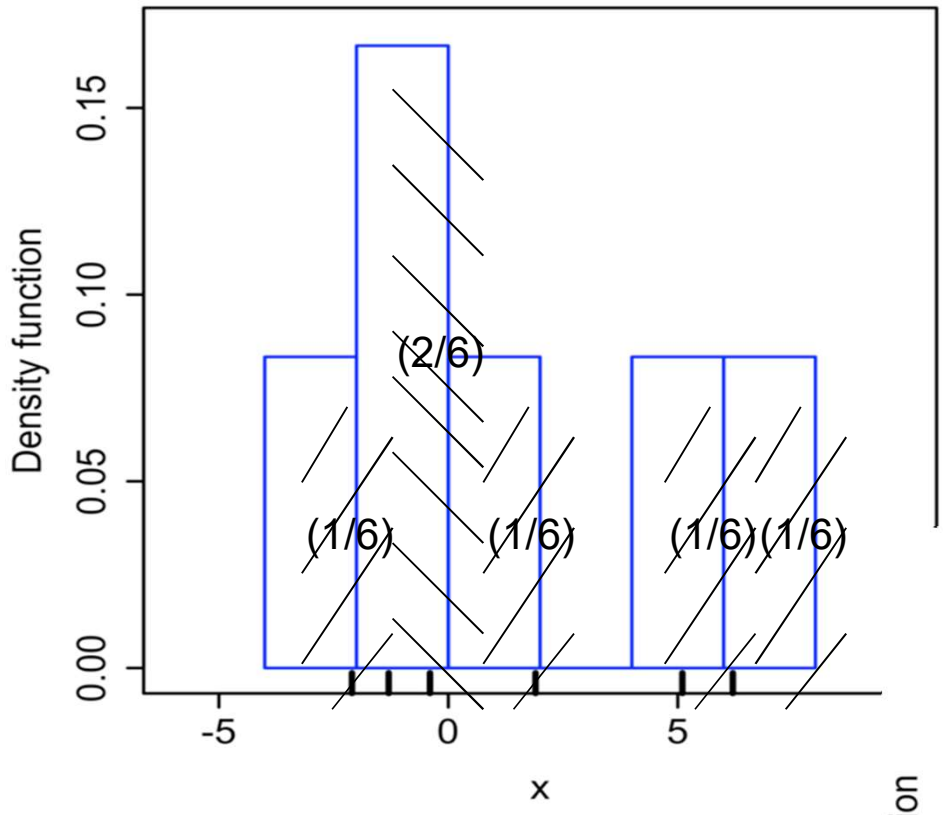
Kernel density estimator

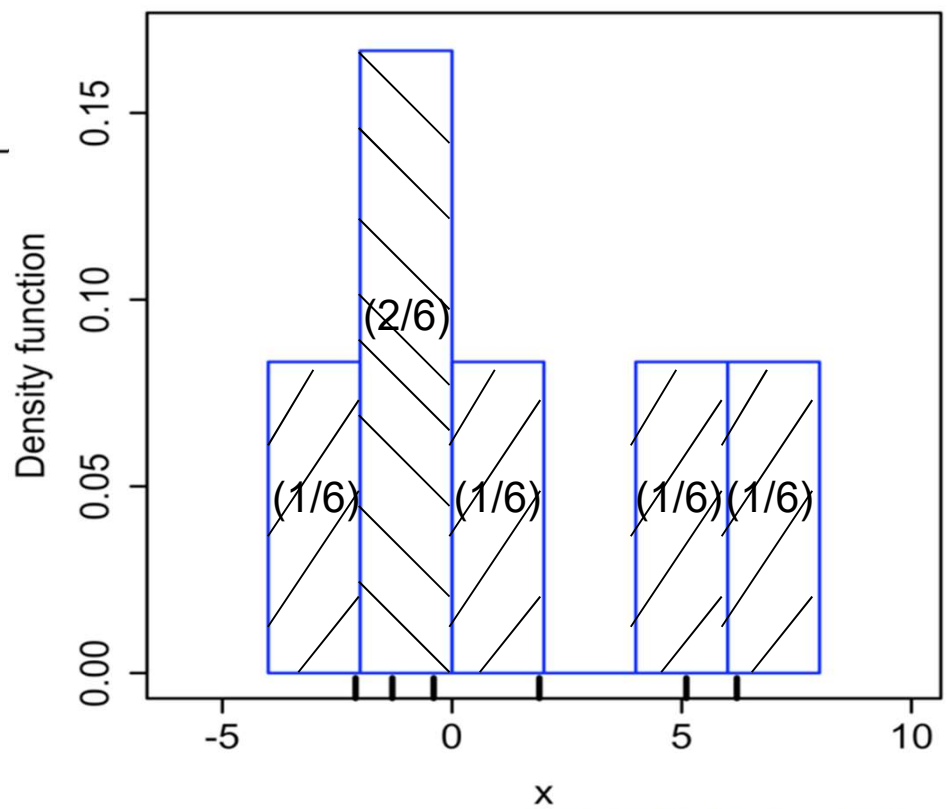
Histogram

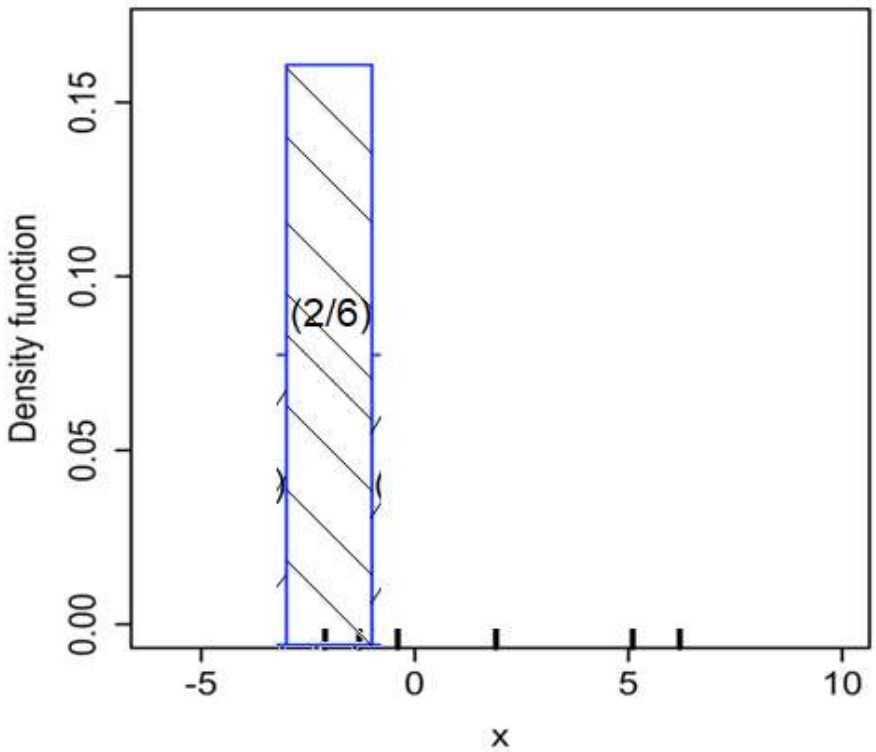
$$P_{estimate} \propto \# \text{ of points in the bin where } x \text{ falls}$$

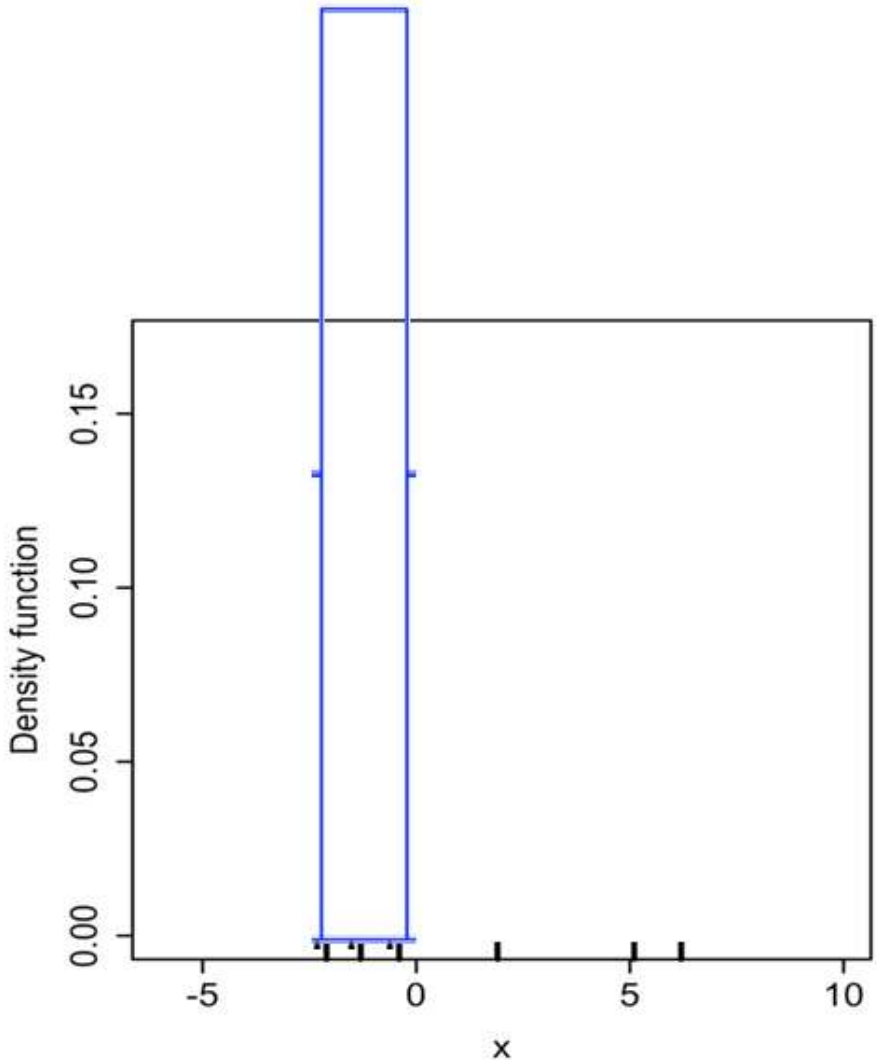
$$P_{estimate} \propto \# \text{ of points in } \mathcal{D} \text{ "near" } x$$

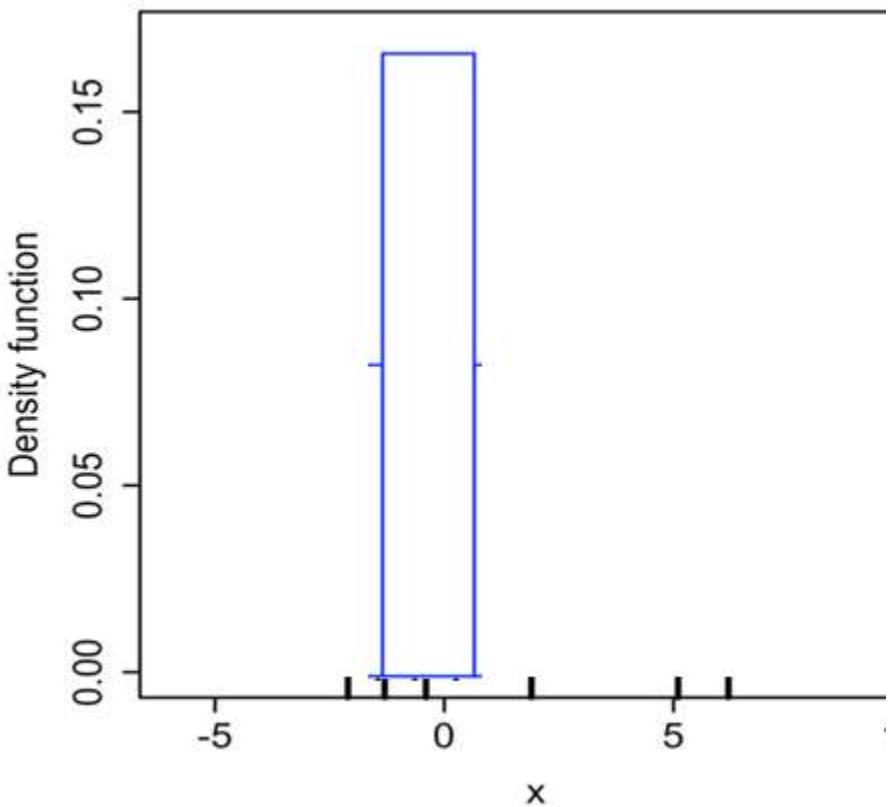
Kernel density

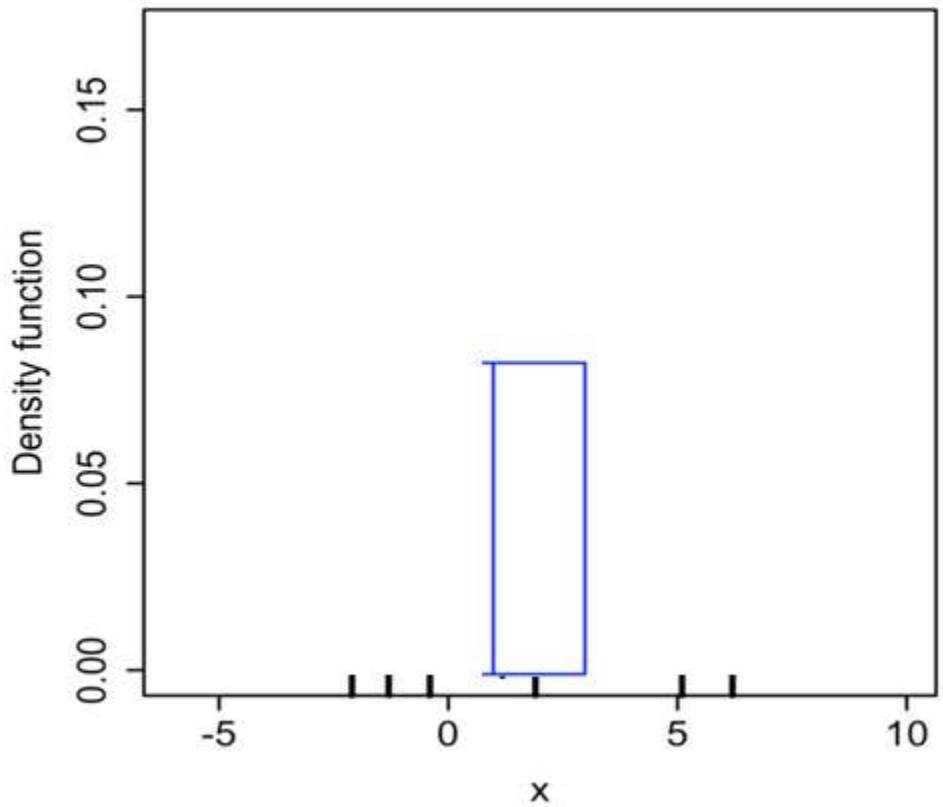

Density function
0.00
0.05
0.10
0.15
(2/6)
(1/6)
(1/6)
(1/6)
(1/6)
-5
0
5
x


Density function
0.00
0.05
0.10
0.15
(2/6)
(1/6)
(1/6)
(1/6)
(1/6)
-5
0
5
10
x


Density function
0.00
0.05
0.10
0.15
(2/6)
-5
0
5
10
x


Density function
0.00
0.05
0.10
0.15
-5
0
5
10
x

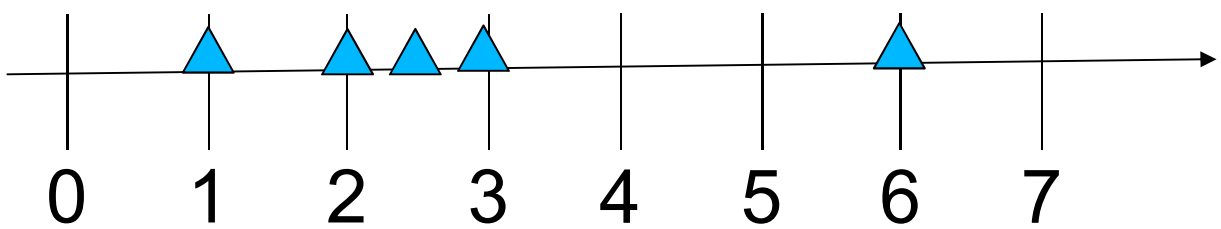

Density function
0.00
0.05
0.10
0.15
-5
0
5
x


Density function
0.00
0.05
0.10
0.15
-5
0
5
10
x

Example

Given a set of five data points x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6

Find **<u>Parzen probability density function</u>** (pdf) estimates at x = 3, using the Gaussian function with σ = 1 as window function



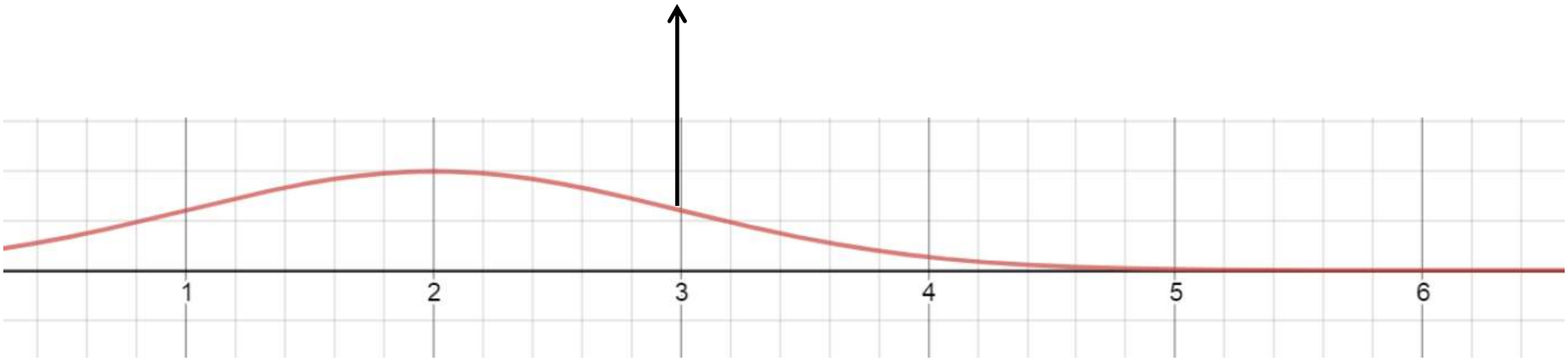
Algorithm

<u>x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6</u>

1. Place a Gaussian at x=2 i.e. μ = 2
2. Find its value @ x=3
3. Place a Gaussian at x=2.5 i.e. μ = 2.5
4. Find its value @ x=3
5. ..
6. ..
7. ..
8. ..
9. Place a Gaussian at x=6 i.e. μ = 6
10. Find its value @ x=3

Gaussian with μ = 2

Find value x=3

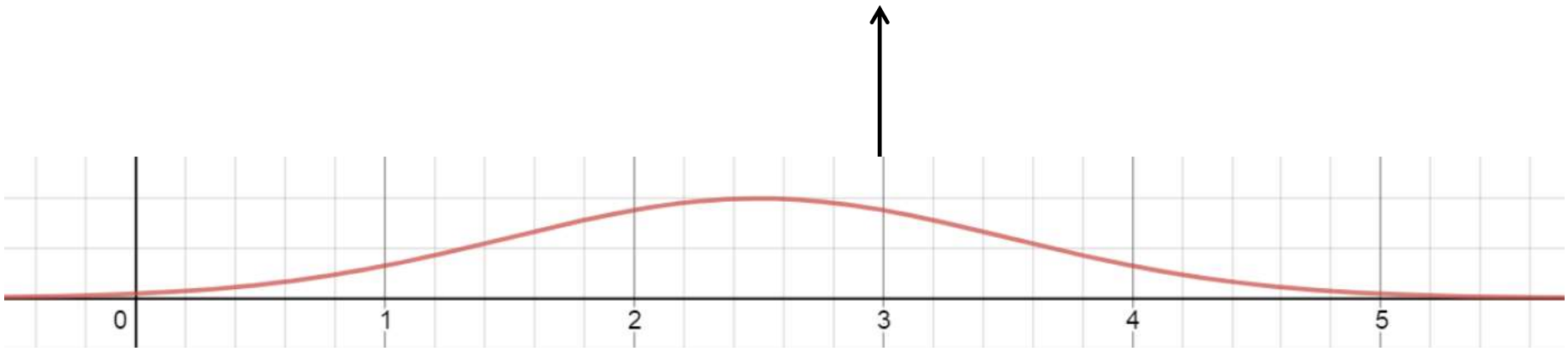


$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_1 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2-3)^2}{2}\right) = 0.2420$$

Gaussian with μ = 2.5

Find value x=3

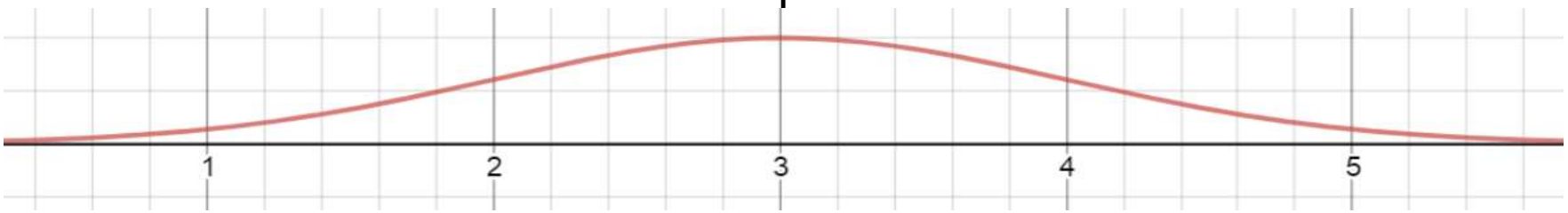


$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_2 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2.5 - 3)^2}{2}\right) = 0.3521$$

Gaussian with μ = 3

Find value x=3 (we get maximum value)

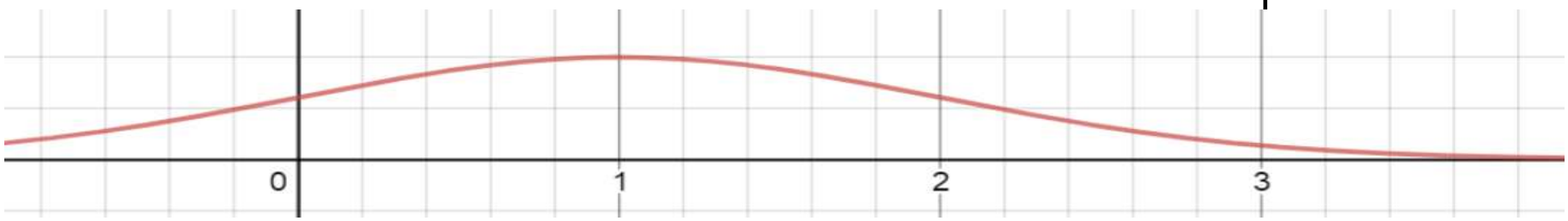


$$= \frac{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_3 - x)^2}{2}\right)}{\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(3-3)^2}{2}\right)} = 0.3521$$

0.3989

Gaussian with μ = 1

Find value x=3



$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_4 - x)^2}{2}\right)$$

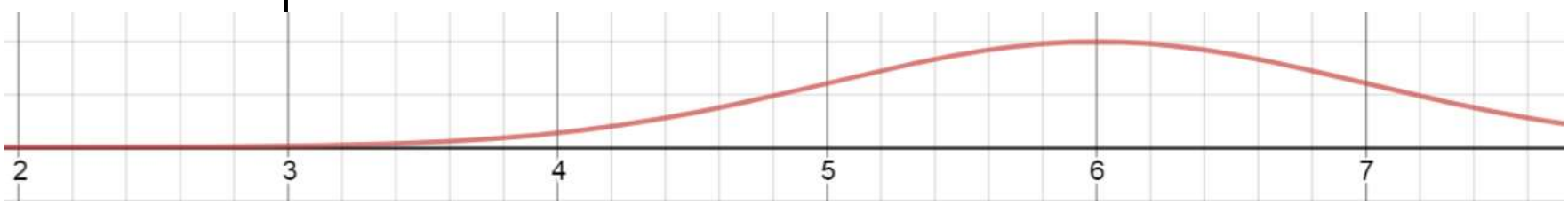
$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2.5 - 3)^2}{2}\right) = 0.3521$$

1

0.054

Gaussian with μ = 6

Find value x=3



$$\frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x_5 - x)^2}{2}\right)$$

$$= \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(2.5 - 3)^2}{2}\right) = 0.3521$$

6

0.0044

- Parzen window is a non-parametric estimator; it doesn't assume a parametric form but uses kernels to "smooth" the observed data.
- Using a Gaussian window creates a smooth estimate for the underlying density function.
- Larger σ leads to more smoothing; small σ produces spikier estimates.
- This method gives a way to estimate the probability density at any point from discrete samples.

Summary Table

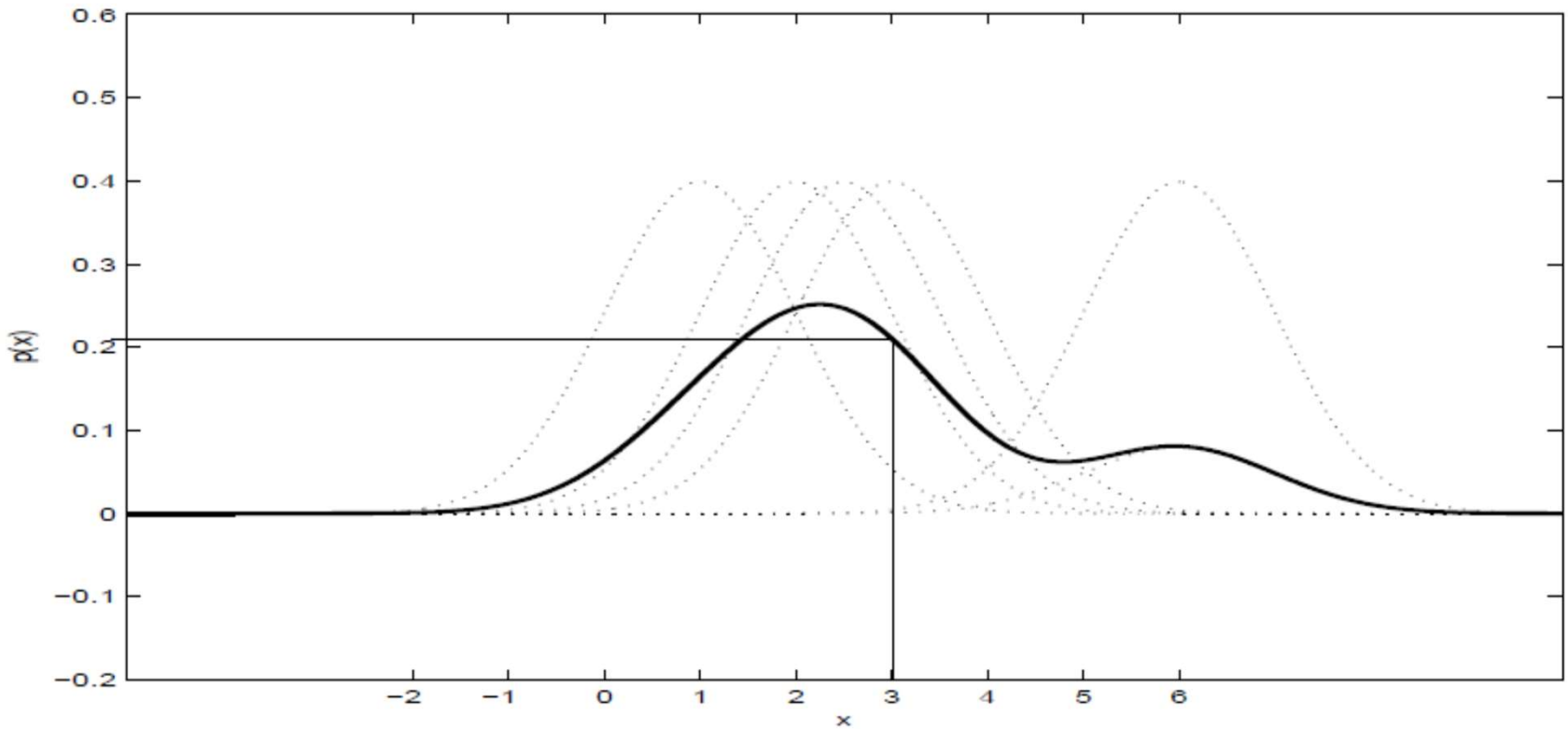
Data point	Value at x=3
x_1=2	≈ 0.2420
x_2=2.5	(calculate similarly)
x_3=3	≈ 0.3989
x_4=1	(calculate similarly)
x_5=6	(calculate similarly)

Parzen estimate at x = 3:

$$p(x=3) = (0.2420 + 0.3521 + 0.3989$$

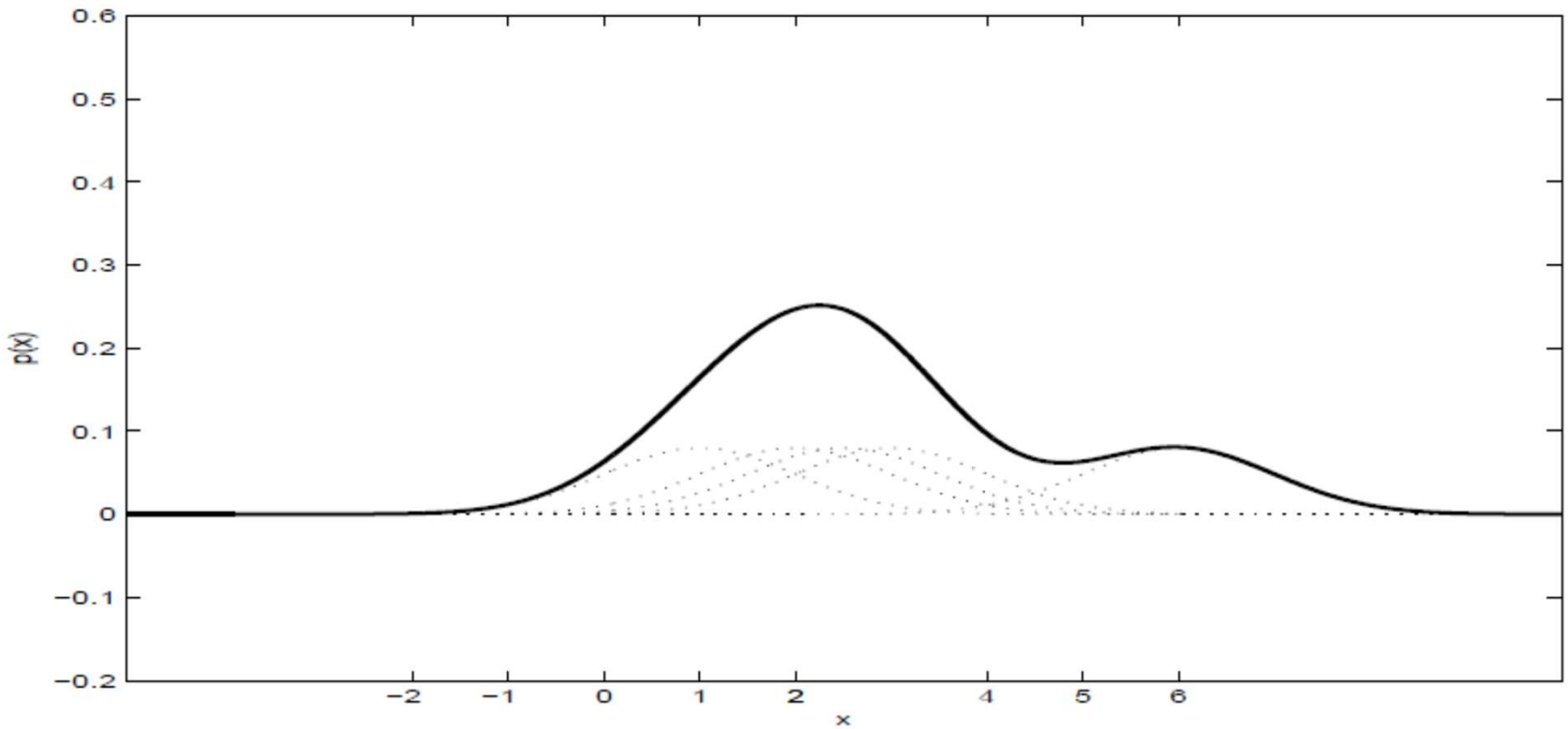
$$+0.0540 + 0.0044)/5 \quad = 0.2103$$

x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6



What is p(3)? Answer is 0.21

Given: x1 = 2, x2 = 2.5, x3 = 3, x4 = 1 and x5 = 6



What is p(x) in general?
Answer is the estimated curve

Algorithm

- n data points $\{X_1, X_2, \ldots, X_n\}$
- Box center is x
- Box width x−h to x+h
- When a data point X_i will fall with in this window?
- If $(|x-X_i|/h) < 1$ then X_i falls within the window otherwise does not fall
  - i.e. all the data points are weighted by a weighting function $w(|x-X_i|/h)$
    - If the argument is <1 then X_i is given weightage of 1/(2nh) otherwise zero

Slide 1: Box (Histogram) Window Algorithm – Step-by-Step

- **n data points** $\{X_1, X_2, \ldots, X_n\}$:

  Represents your observed data.
- **Box center is x:**

  We want the estimated density at a particular value x .
- **Box width from x-h to x+h:**

  The window (kernel) is a box (rectangle) of width $2h$, centered at x.
- **Determine if data point is in the window:**

  A data point X_i is inside the window if its distance from x, normalized by h, is less than 1:

$$\left|\frac{x - X_i}{h}\right| < 1$$

  This checks if X_i falls within $[x-h, x+h]$.
- **Weight assignment:**

  If X_i is inside the box, it counts as a "hit" (gets weight), otherwise its weight is zero. All hits contribute equally, and their weight is

$$w = \frac{1}{2nh}$$

which depends on the total number of data points n and window width h .

Slide 2: Formal Mathematical Expression

The Parzen estimator using the box (uniform) window is given by:

$$\hat{f}(x) = \frac{1}{nh}\sum_{i=1}^{n} w\left(\frac{x - X_i}{h}\right)$$

Where the weight function w(x) defines the shape of the window:

$$w(x) = \begin{cases} \frac{1}{2} & \text{if } |x| < 1 \\ 0 & \text{otherwise} \end{cases}$$

Interpretation:

- For each data point, you see if it falls inside the window ($|x-X_i| < h$). If yes, its weight is $1/(2nh)$, otherwise zero.
- Summing for all data points gives the estimated pdf at point x .

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} w\left(\frac{x - X_i}{h} \right)$$

$$w(x) = \frac{1}{2} \quad if \quad |x| < 1$$

$$= 0 \quad otherwise$$

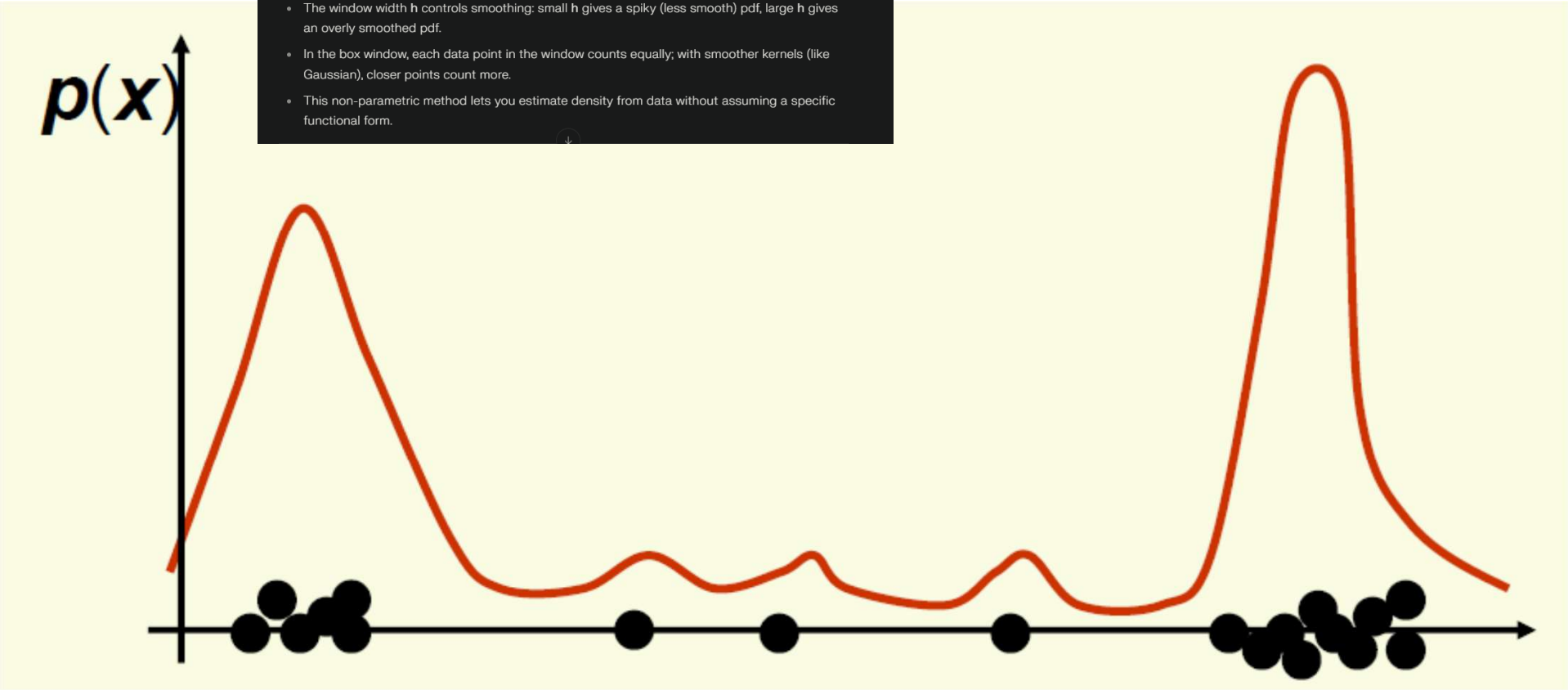
Answer Sources 3

Summary Table

Concept	Plain English Description
Box Window	Count points in [x-h, x+h]; each contributes equally
Weight Function	w = 1/(2nh) if within window; zero otherwise
Density Estimation	For every x, sum contributions and normalize
Parzen Curve Output	Smoothed version of histogram, capturing local density

Key Points

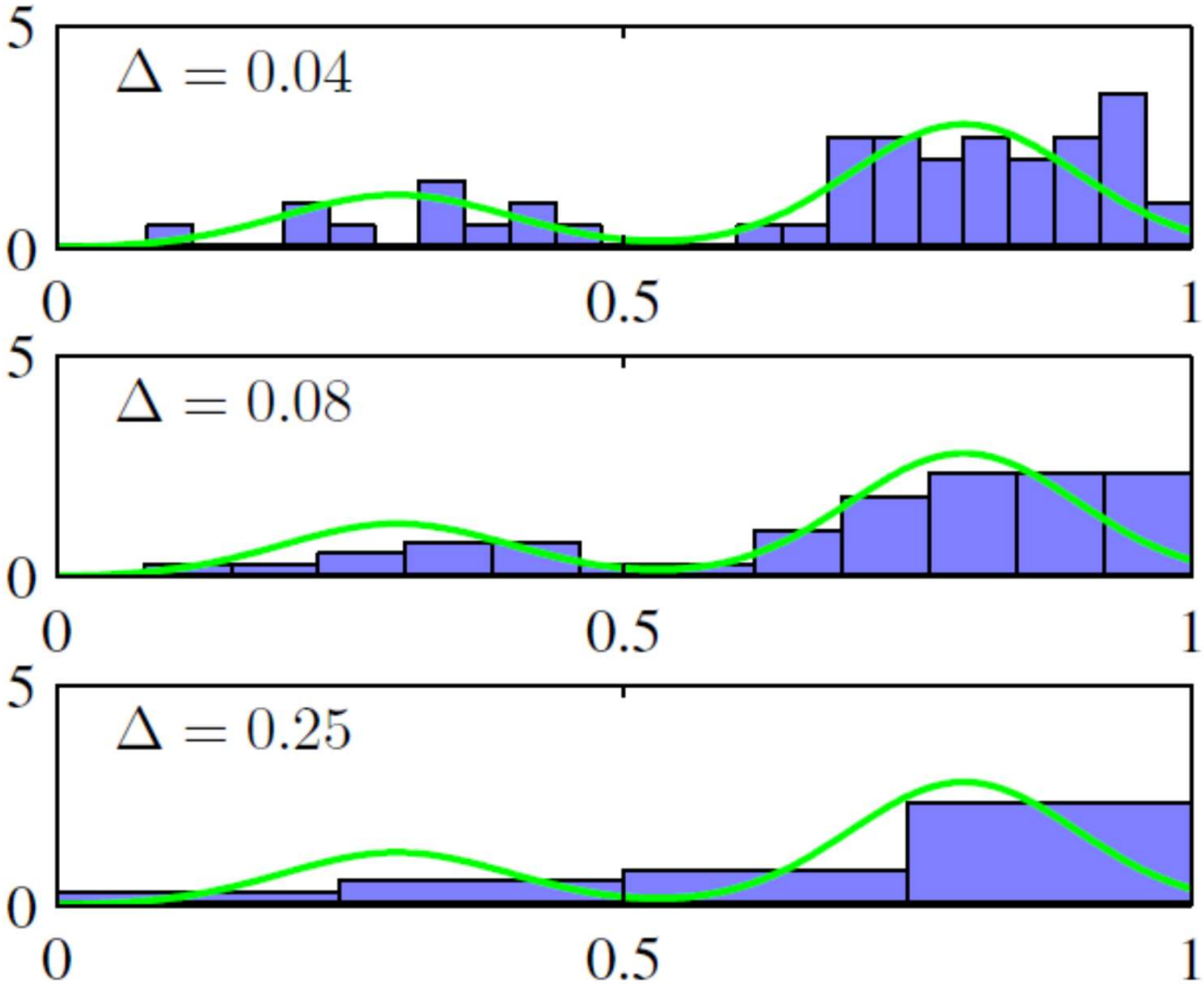
- The window width **h** controls smoothing: small **h** gives a spiky (less smooth) pdf, large **h** gives an overly smoothed pdf.
- In the box window, each data point in the window counts equally; with smoother kernels (like Gaussian), closer points count more.
- This non-parametric method lets you estimate density from data without assuming a specific functional form.



In essence:

The box (histogram/rectangular) kernel Parzen window method estimates density at a point by counting how many data points fall inside a fixed window around that point, assigns them uniform weight, and normalizes. The result is a flexible, intuitive approach for discovering the underlying pdf of your data.

Impact of Bin width (Δ)



- Green is the correct distribution
- When Δ = 0.04 or 0.25, Histogram do not reflect the green

Density estimate

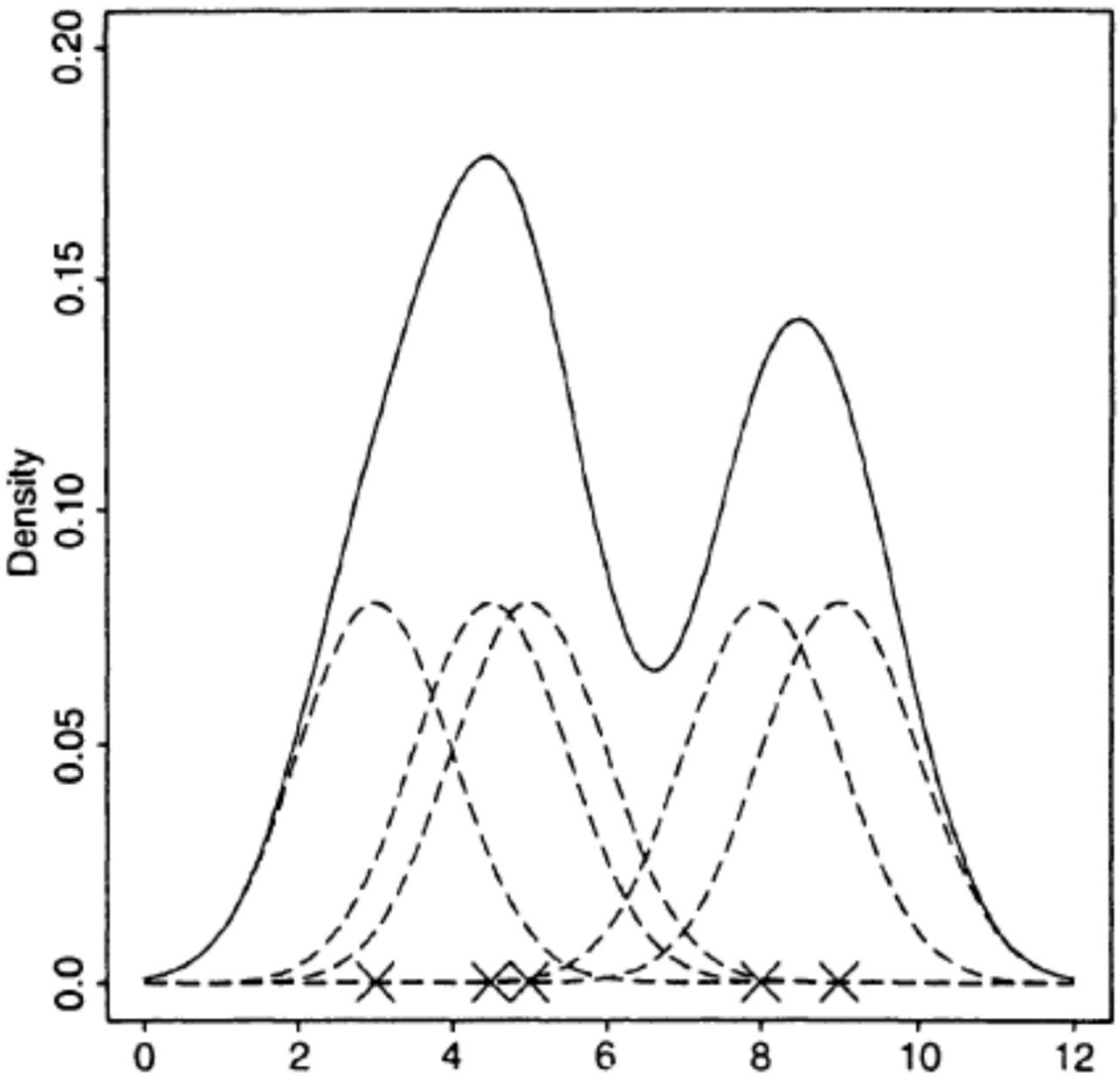
- Imagine continuous distribution
- We got certain points
- Interested in estimating the distribution of data from the given data

Kernel estimator

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} k\left(\frac{x - X_i}{h} \right)$$

Parzen window estimator: Use normal Gaussian

Kernel density estimate based on five observations

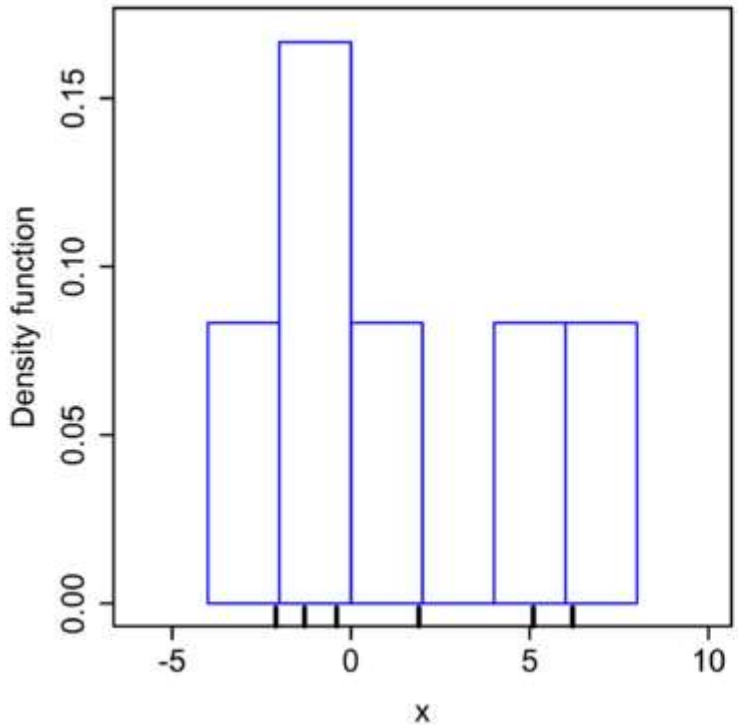


Parzen Window

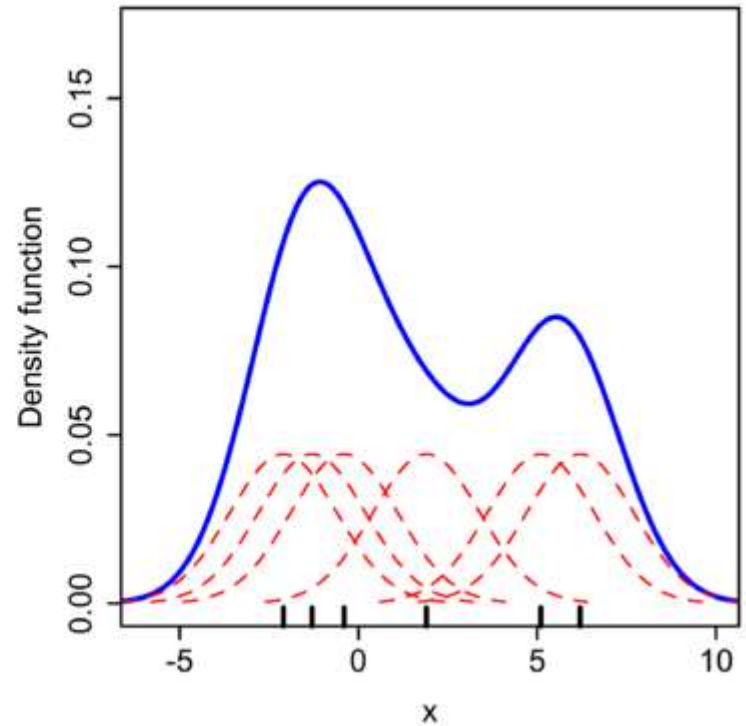
- Number of data points (n)
- n Gaussian computations to calculate the density @ a point
- To find density @ n points we need n x n i.e. n^2 calculations
- Instead of 1 dimensional data if we have d d dimensional then we need d times more computations.

Sample No.	Value
1	-2.1
2	-1.3
3	-0.4
4	1.9
5	5.1
6	6.2

Histogram-based Density



Parzen window-based Density



Histogram changed to KDE → Tophat KDE

- Bin edge is no more fixed by the user
- Bin is placed centered around data points

Key Points

- **Histogram** is simple and fast, but can be misleading if bin edges are chosen poorly.
- **Parzen/KDE** offers more accurate and nuanced density estimation, but at higher computational cost (especially with many data points or dimensions).
- KDE methods (including Parzen window) are preferable for most modern data analysis, especially if visualization or downstream modeling benefits from smooth density curves.

In summary:

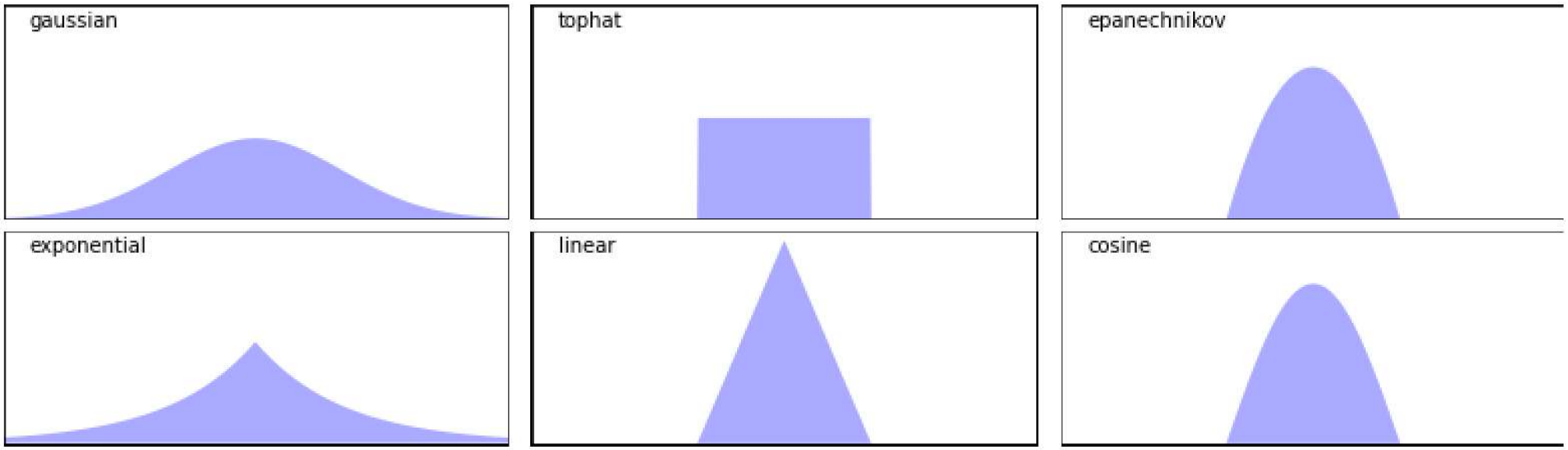
Parzen window (KDE) advances histogram-based density estimation by centering kernels on data points, using all points for every density computation, and producing a smooth estimate. While computationally heavier, especially for large and high-dimensional data, it delivers superior and less arbitrary representation of your data's true underlying distribution.

Mathematical Formula image.jpg

$$\hat{f}(x) = \frac{1}{nh} \sum_{i=1}^{n} k\left(\frac{x - X_i}{h}\right)$$

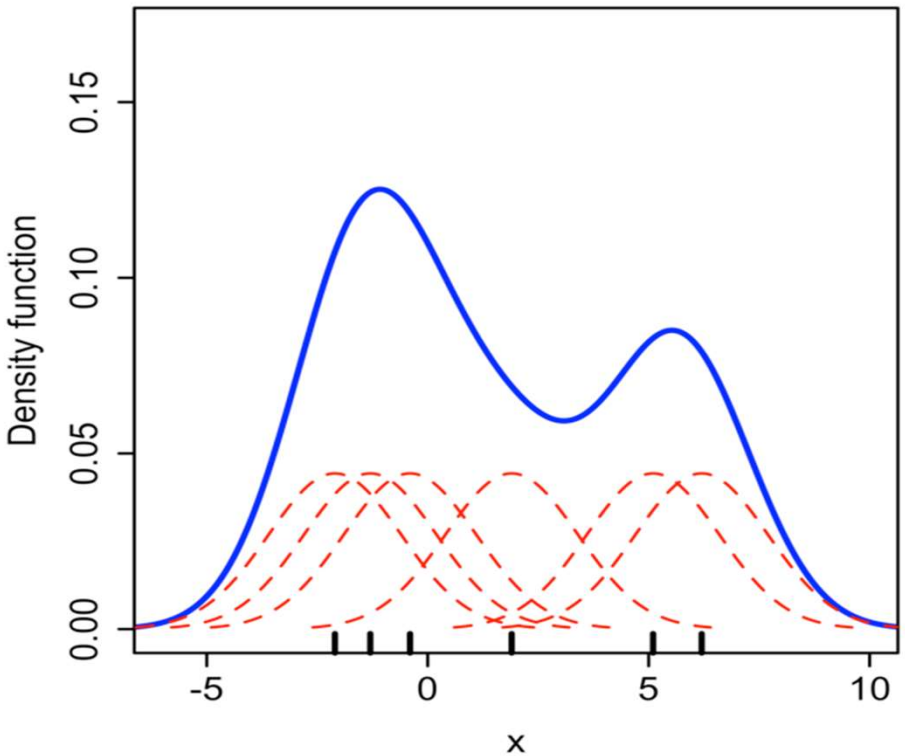
- $\hat{f}(x)$: Estimated probability density at point x
- n: Number of data points
- X_i: The i-th observed data point
- h: Bandwidth (controls smoothing)
- $k(\cdot)$: The kernel function, often a Gaussian (normal) curve

Choice of kernels



Kernel density estimation

- Normal kernel with variance 2.25 on each of the data points x_i
- Kernels are summed to make the kernel density estimate

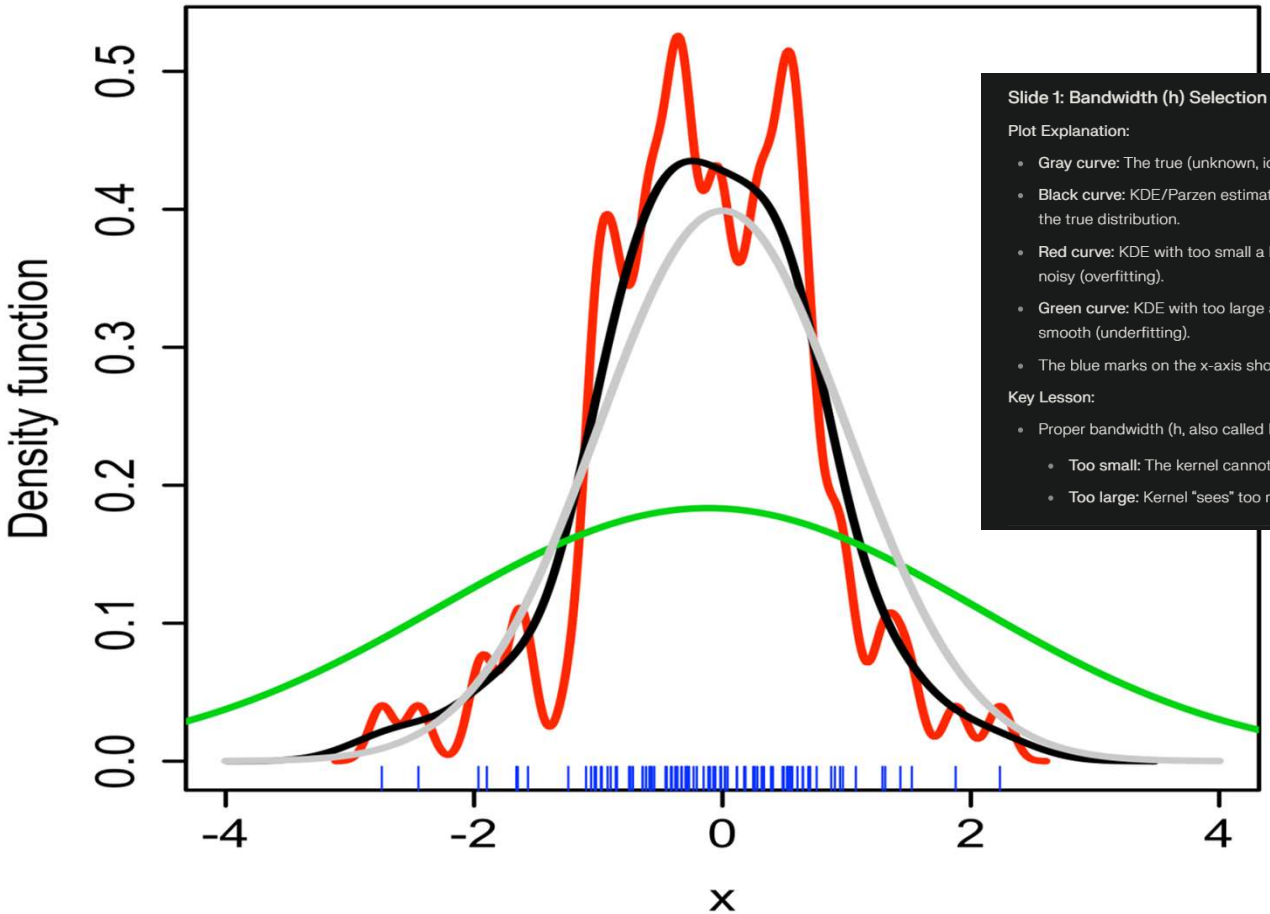


Bandwidth

Variance/bandwidth of kernel – free parameter

Important parameter – decides the smoothness of estimation

Gray: True distribution
Black: appropriate estimation
red: too small bandwidth
green: too high bandwidth



Slide 1: Bandwidth (h) Selection and its Impact

Plot Explanation:

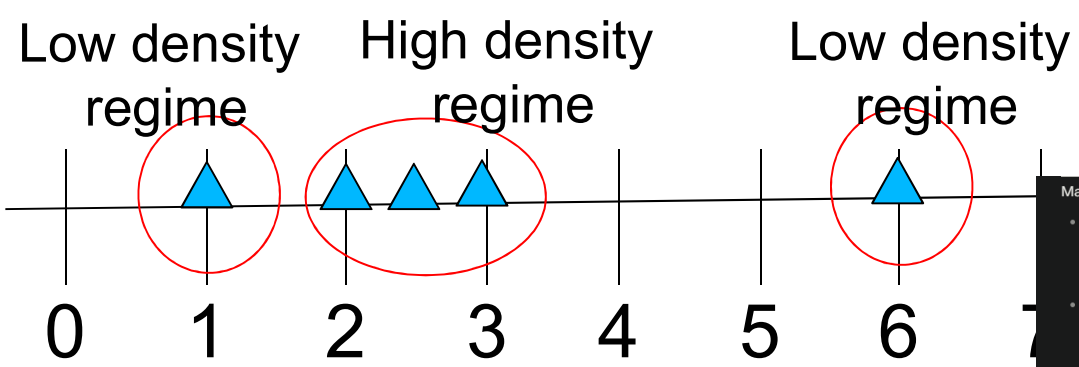
- **Gray curve:** The true (unknown, ideal) underlying data distribution.
- **Black curve:** KDE/Parzen estimate using an appropriate/"just right" bandwidth—very close to the true distribution.
- **Red curve:** KDE with too small a bandwidth (h too small)—the estimate is too "spiky" and noisy (overfitting).
- **Green curve:** KDE with too large a bandwidth (h too large)—the estimate is too "flat" and smooth (underfitting).
- The blue marks on the x-axis show the locations of sample points.

Key Lesson:

- Proper bandwidth (h, also called kernel width or smoothing parameter) is critical:
  - **Too small:** The kernel cannot "see" enough points; result is bumpy/noisy.
  - **Too large:** Kernel "sees" too many points; real structure is lost.

Parzen density estimation - problems

- kernel width (h) is fixed in all regimes
  - High density regime
  - Low density regime
- Large h in high density regime – over smoothing
- Low h in low density region – noisy estimates



Main Points:

- **Kernel width (h) is fixed:**
  In classic Parzen estimation, the bandwidth h is the same everywhere, regardless of how densely or sparsely data are distributed.
- **Two types of regions:**
  - **High-density regime:** Many data points close together.
  - **Low-density regime:** Data points are sparse/far apart.

Problems:

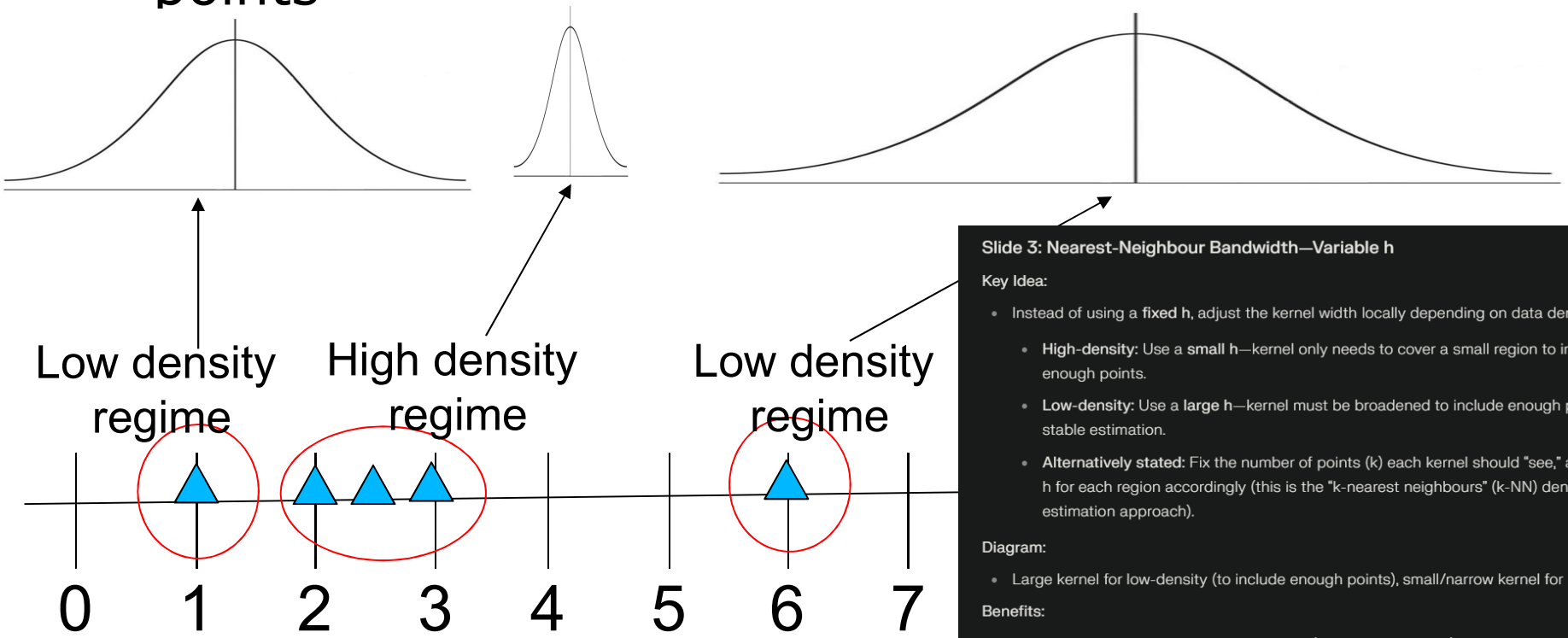
- **Large h in high-density area:**
  Over-smoothing—local features are averaged away, peaks get blurred together.
- **Small h in low-density area:**
  Not enough points fall under the kernel, resulting in a noisy, unreliable estimate.

Diagram:

- On the number line, triangles show where data points are.
- High-density region (lots of points close together) vs. low-density regions on the sides (points are more isolated).

Nearest-neighbour methods

- kernel width (h) NOT fixed
  - High density regime – low h
  - Low density regime – high k
- In other words – choose h to accommodate fixed points



Slide 3: Nearest-Neighbour Bandwidth—Variable h

Key Idea:

- Instead of using a **fixed h**, adjust the kernel width locally depending on data density.
  - **High-density:** Use a **small h**—kernel only needs to cover a small region to include enough points.
  - **Low-density:** Use a **large h**—kernel must be broadened to include enough points for stable estimation.
  - **Alternatively stated:** Fix the number of points (k) each kernel should "see," and choose h for each region accordingly (this is the "k-nearest neighbours" (k-NN) density estimation approach).

Diagram:

- Large kernel for low-density (to include enough points), small/narrow kernel for high-density.

Benefits:

- Reduces over-smoothing in dense regions (can preserve detail).
- Reduces noise in sparse regions (improves reliability of the estimate).